

CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

- Energy bands in conductors, semiconductors and insulators (qualitative ideas only)
- Intrinsic and extrinsic semiconductors - p and n type
- p-n junction Semiconductor diode
- I-V characteristics in forward and reverse bias
- Application of junction diode - diode as a rectifier.

Classification of Metals, Insulator and Semiconductor

Semiconductor Electronics: Material Devices and Simple circuit

ON THE BASIS OF CONDUCTIVITY

(1) For metals:

$$S \sim 10^2 - 10^8 \Omega m$$
$$\sigma \sim 10^2 - 10^8 S/m$$

They have high conductivity.

(2) For Semiconductors:

$$S \sim 10^{-5} - 10^6 \Omega m$$
$$\sigma \sim 10^5 - 10^6 S/m$$

They have intermediate conductivity to metals and insulators.

(3) For insulators:

$$S \sim 10^{11} - 10^{19} \Omega m$$
$$\sigma \sim 10^{-11} - 10^{-19} S/m$$

They have low conductivity

σ = electrical conductivity

ρ = resistivity

SEMICONDUCTOR AND ITS TYPES

Semiconductors exhibit electrical conductivity between conductors and Non-Conductors.

INTRINSIC SEMICONDUCTORS

- (1) Pure Semiconductors are intrinsic Semiconductors.
- (2) $N_i = N_e = N_h$, where, N_e = No. of free electrons, N_h = No. of holes, N_i = intrinsic carrier Concentration
- (3) Examples:- Ge, Si

EXTRINSIC SEMICONDUCTORS

- (1) Impure or doped Semiconductors are said to be extrinsic Semiconductors
- (2) Impurities are added to improve Conductivity

N - type Semiconductor $N_e \gg N_h$

- (1) Electrons are majority charge carriers.
- (2) Holes are minority charge carriers.
- (3) Si or Ge doped with Pentavalent elements (P, As, Sb)

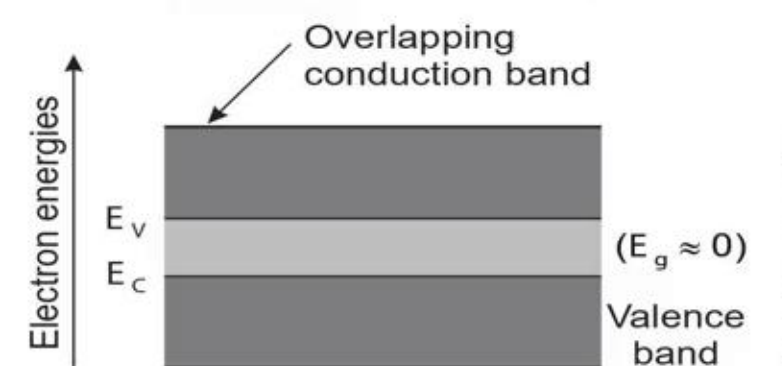
P - type Semiconductor $N_h \gg N_e$

- (1) Si or Ge doped with trivalent (B, Al) elements
- (2) Electrons are minority charge carriers.
- (3) Holes are majority charge carriers

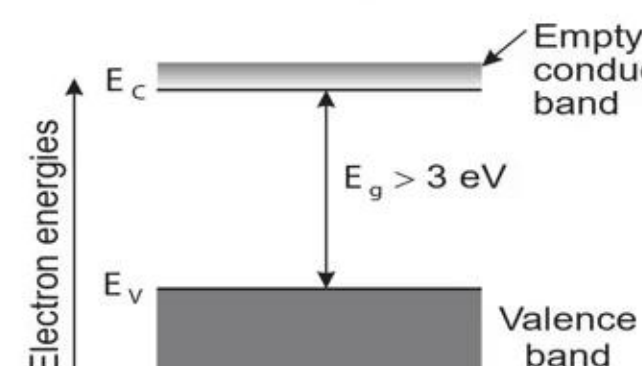
THERMAL EQUILIBRIUM

The electron and hole Concentration in a Semiconductor in thermal equilibrium is given by, $n_e n_h = n_i^2$

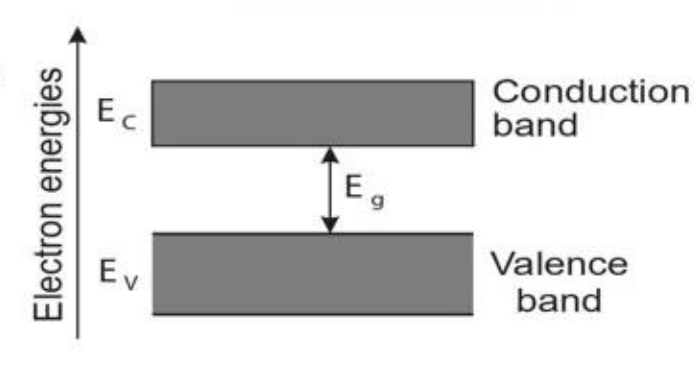
CONDUCTOR (METAL)



INSULATOR



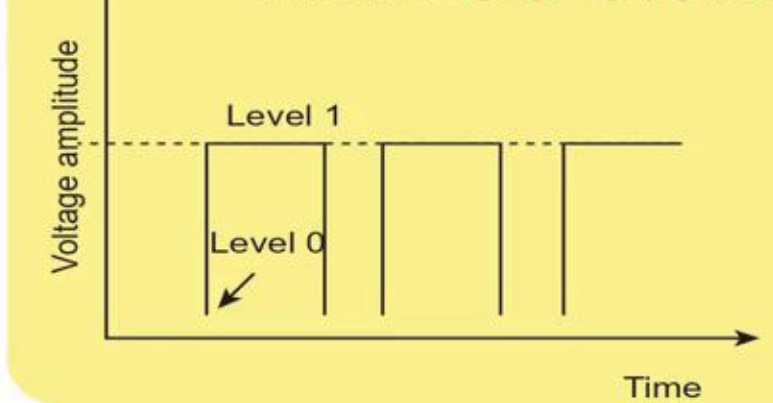
SEMICONDUCTOR



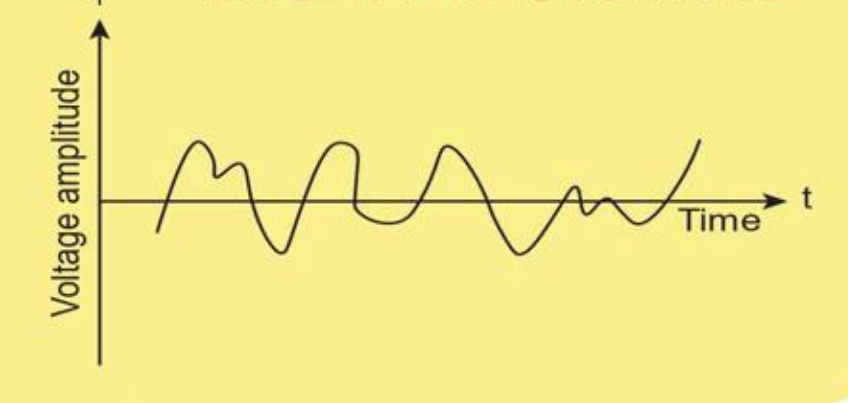
TRANSISTOR

Transistor is a three terminal device
(1) Emitter (E)
(2) Base (B)
(3) Collector (C)

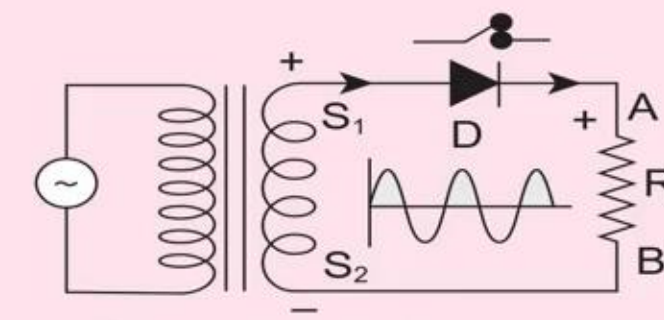
ANALOG SIGNAL



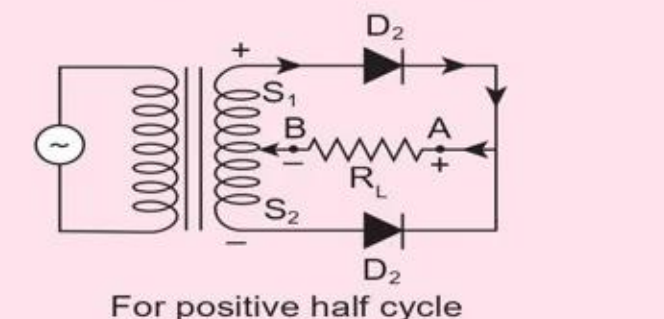
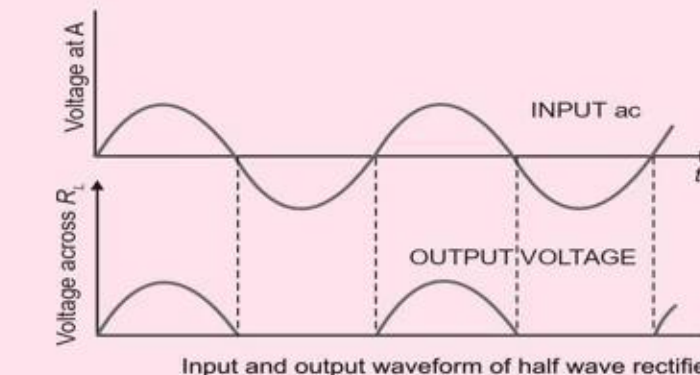
DIGITAL SIGNAL



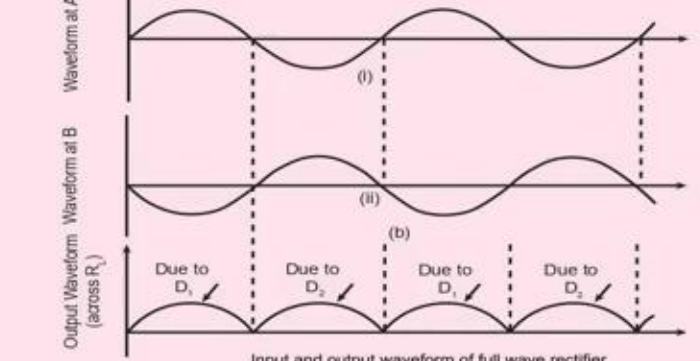
APPLICATIONS OF JUNCTION DIODE



For positive half cycle



For positive half cycle



P - N JUNCTION DIODE

P - N junction diode is the combination of P - type and N - type Semiconductor.
P - region has mobile majority holes and immobile - ve ions.
N - region has mobile majority free electrons and immobile positively charged ions.

POTENTIAL BARRIERS

Potential barrier is the potential difference developed across depletion region.

$$V_B = 0.7 \text{ for Silicon}$$
$$= 0.3 \text{ for germanium}$$

FORWARD BIAS

In Forward Bias

- (1) +ve terminal to P - Side
- (2) -ve terminal to N - Side
- (3) depletion layer reduced
- (4) diffusion current increases

REVERSE BIAS

In Reverse Bias

- (1) -ve terminal to P - Side
- (2) +ve terminal to N - Side
- (3) depletion layer increases
- (4) diffusion current increases

ZENER BREAKDOWN

- This phenomenon takes place in
- (1) P - N junction having 'high doping'
 - (2) P - N junction having thin depletion layer
- Here, P - N junction does not damage permanently

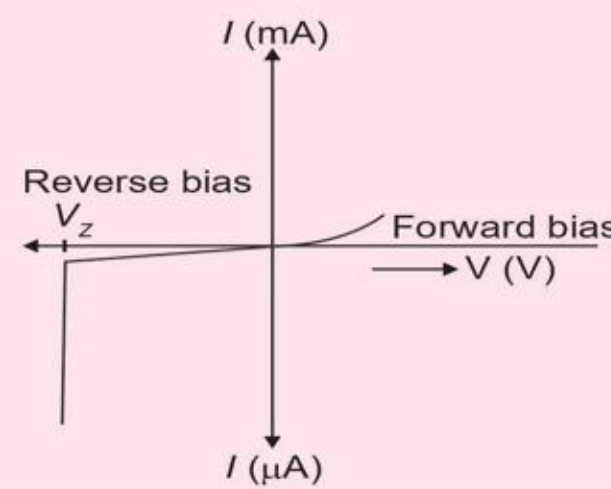
AVALANCHE BREAKDOWN

- This phenomenon takes place in
- (1) P - N junction having 'low doping'
 - (2) P - N junction having thick depletion layer.
- Here, P - N junction damages permanently due to abruptly increment of minorities during repetitive collisions.

SEMICONDUCTOR ZENER DIODE DIODES

Anode(A) Cathode(K)
Zener Diode

I - V characteristics

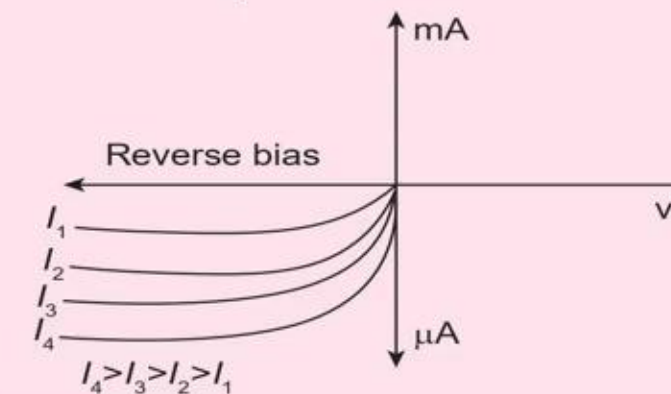


PHOTODIODE

The symbol of photodiode is

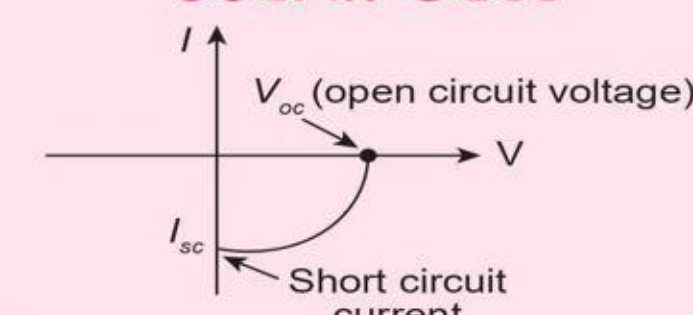


I - V characteristics of a photodiode

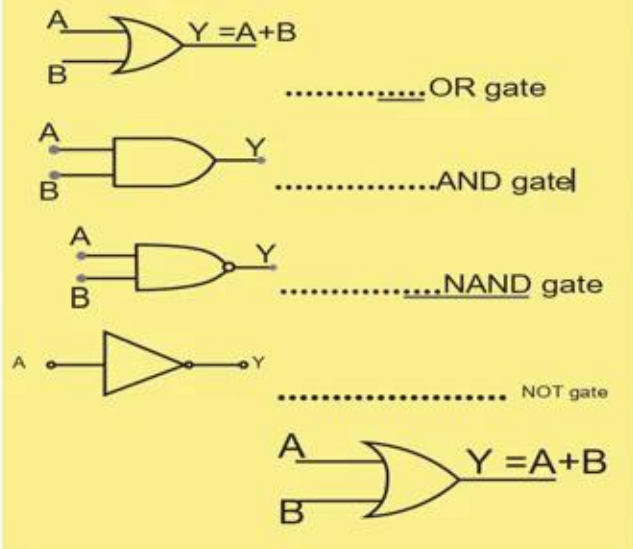


I - V CHARACTERISTICS OF A SOLAR CELL.

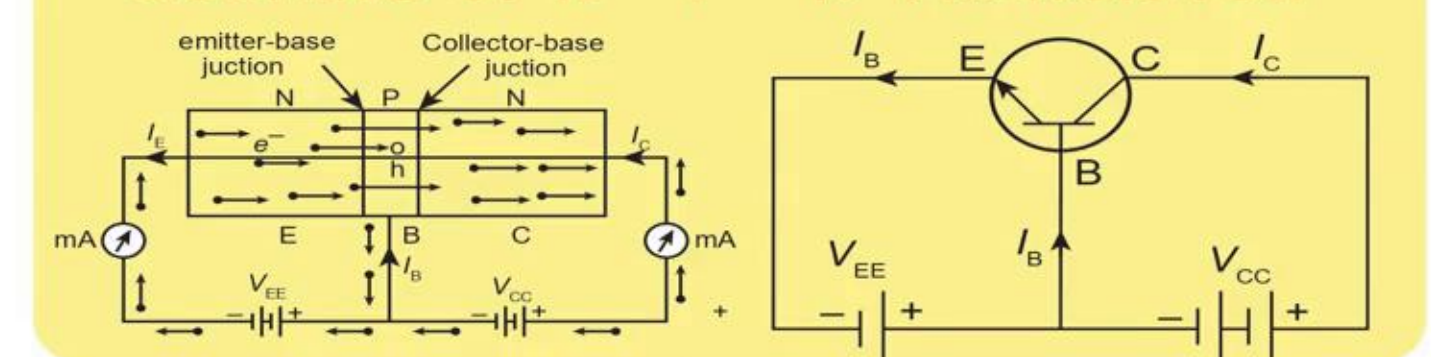
SOLAR CELL



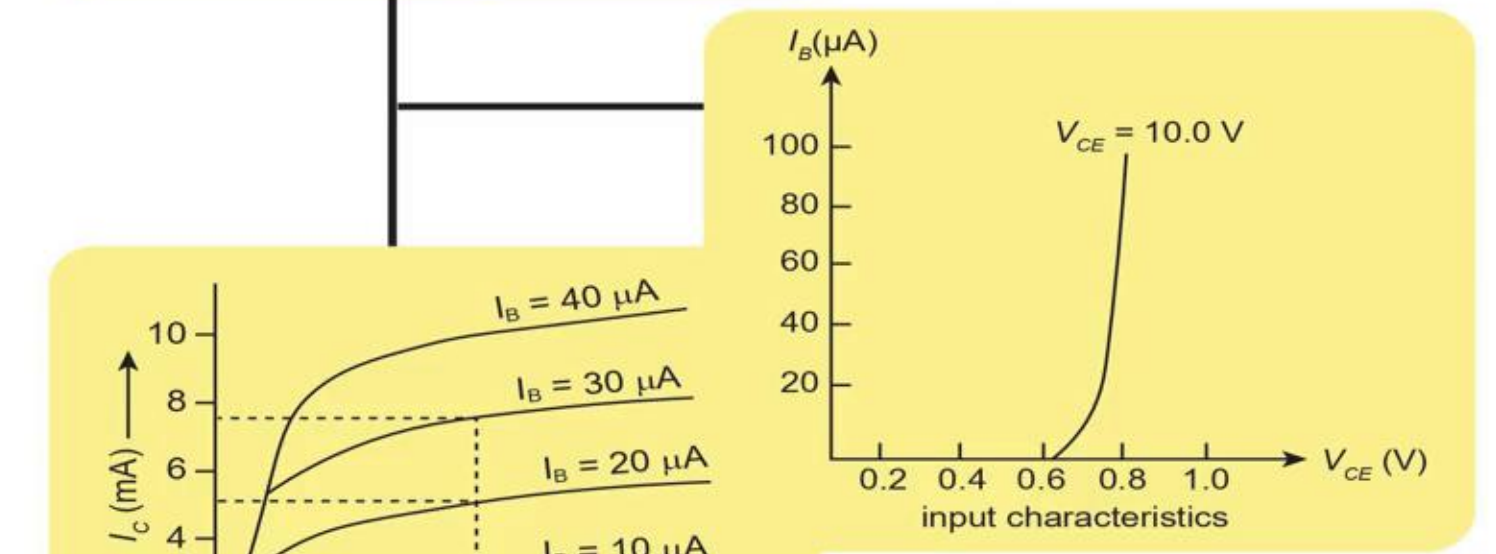
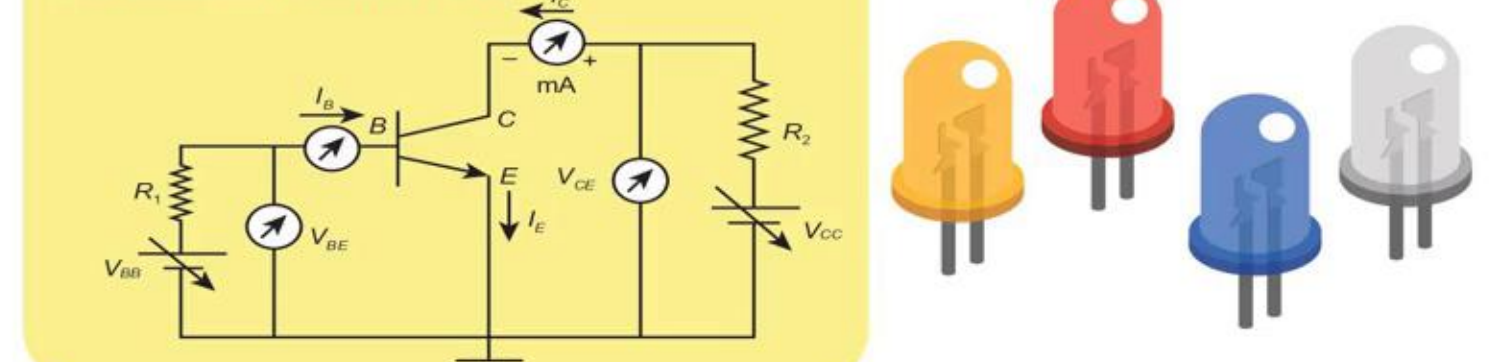
BASIC LOGIC GATES



WORKING OF N - P - N TRANSISTOR

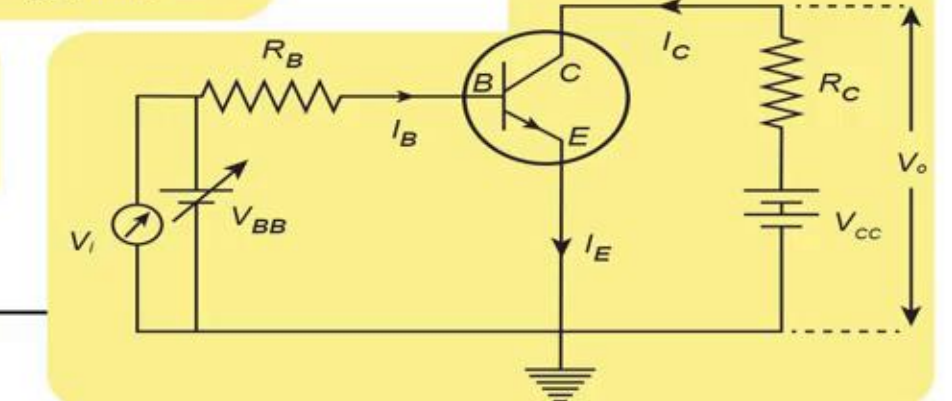


CE CONFIGURATION CHARACTERISATION

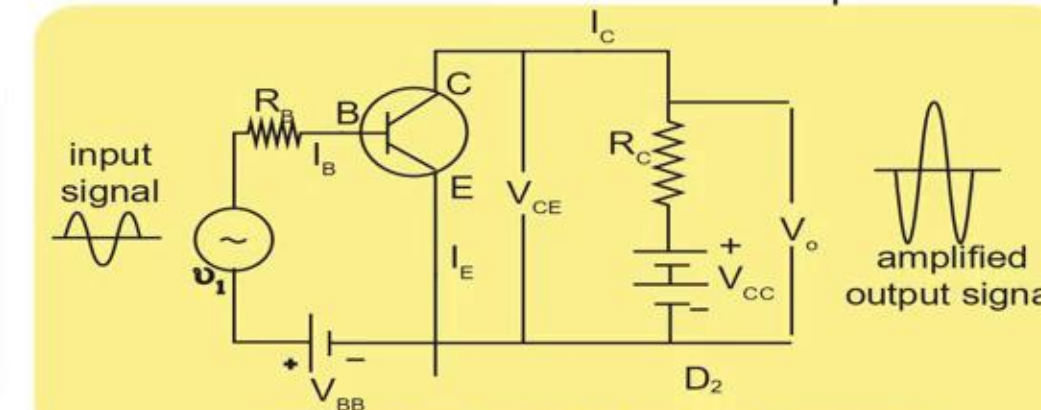
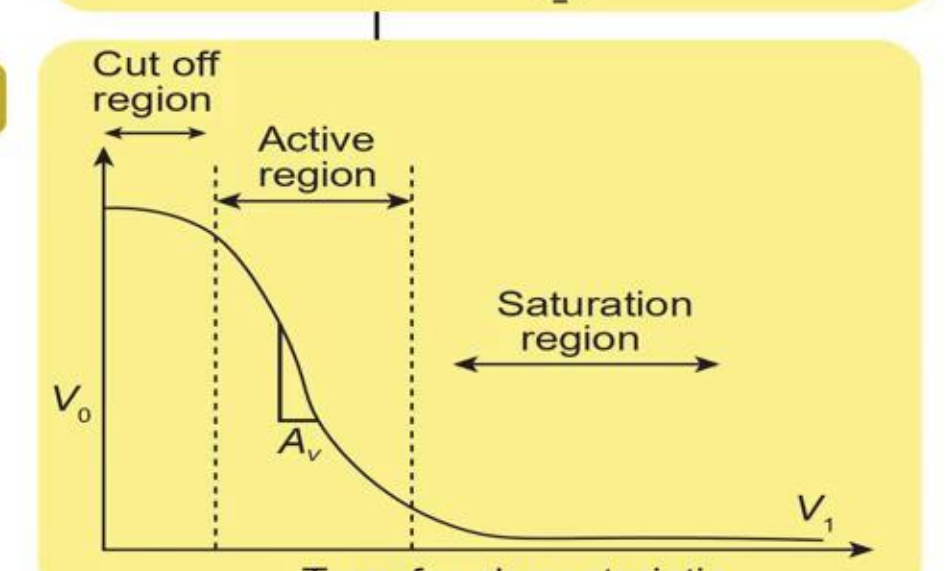


APPLICATION OF TRANSISTOR

AS A SWITCH

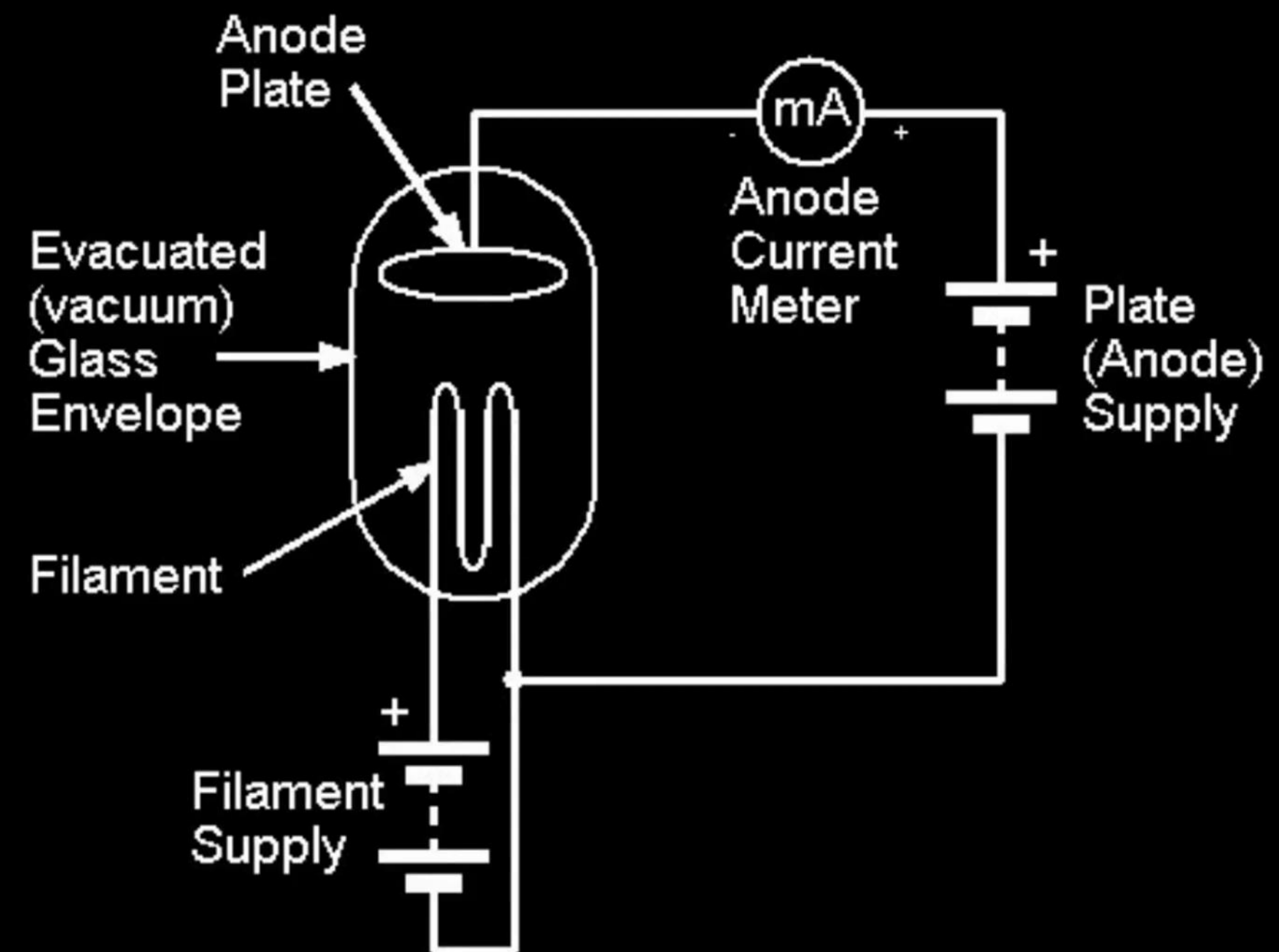
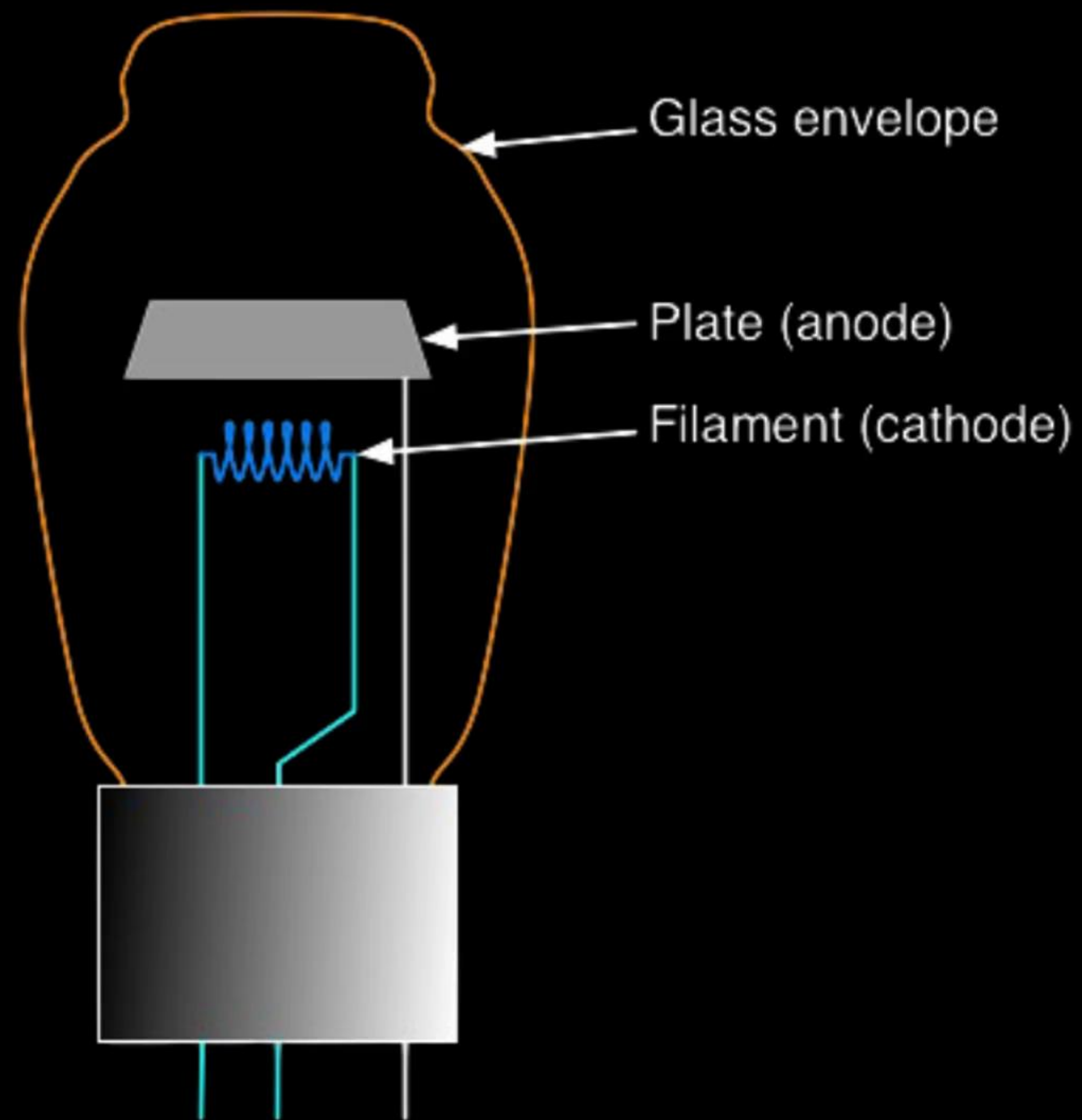


AS AN AMPLIFIER



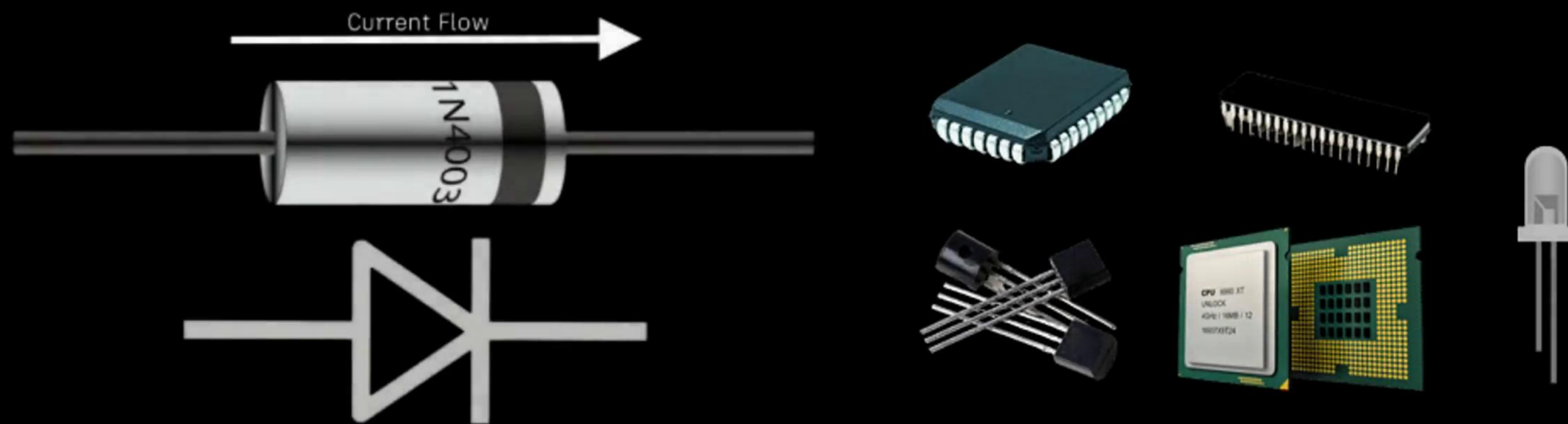
Introduction

- Before 1948, **vacuum tubes (valves)** like diode and triode were used as electronic devices.



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- Vacuum tubes were **bulky, consumed high power, required high voltages, and had low reliability.**
- **Semiconductors** (discovered in 1930s) allowed control of charge carriers inside solids without vacuum or heating.



Introduction

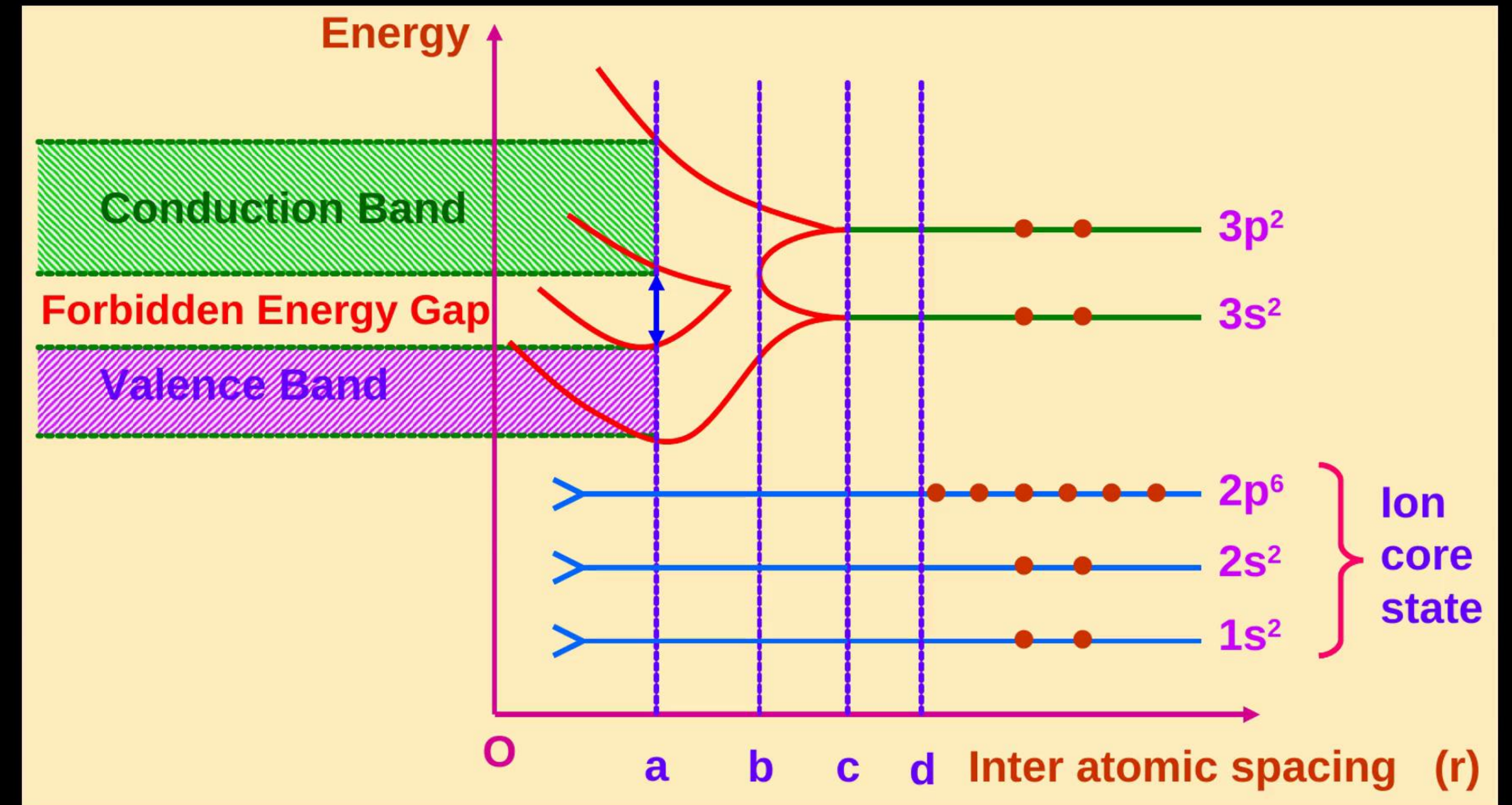
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- Vacuum tubes were **bulky, consumed high power, required high voltages, and had low reliability.**
- **Semiconductors** (discovered in 1930s) allowed control of charge carriers inside solids without vacuum or heating.
- Advantages of semiconductors: **small size, low power, low voltage operation, long life, high reliability.**
- This chapter introduces **semiconductor physics, diodes, A few circuits illustrating their applications.**

Classification of Metals, Conductors and Semiconductors

- **Metals:** Very low resistivity (10^{-2} - $10^{-8} \Omega m$) and high conductivity (10^2 - $10^8 S m^{-1}$).
- **Insulators:** Very high resistivity (10^{11} - $10^{19} \Omega m$) and very low conductivity.
- **Semiconductors:** They have resistivity or conductivity intermediate to metals and insulators. (10^{-5} - $10^6 \Omega m$). Conductivity varies with **temperature, light, doping**.
- Types of semiconductors: **Elemental (Si, Ge), Compound - Inorganic (CdS, GaAs, CdSe, InP, etc.), Organic (anthracene), and Polymers (polyaniline, polypyrrole)**.
- Order of conductivity: **Metals $\gg$ Semiconductors $\gg$ Insulators**.

Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.
- Discrete levels split into many closely spaced levels forming **energy bands**.
- Core electrons remain unaffected, only outer (valence) electrons form bands.
- Consider Si having Electronic configuration $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$. (Atomic No. is 14)
- In Si/Ge crystal having N atoms.: $4N$ valence electrons $\Rightarrow 8N$ states split into 2 bands separated by **energy gap** E_g .



Valence Band, Conduction Band & Band Gap

- **Valence Band (VB):** The energy band which includes the energy levels of the valence electrons (**bounded electrons**) is called the valence band. **It is completely filled at 0 K.**
- **Conduction Band (CB):** Band above VB which contains free electrons; **electrons here are free to conduct** is called conduction band. It is normally empty at 0 K.
- **Energy Gap E_g :** Difference between top of VB (E_V) and bottom of CB (E_C).

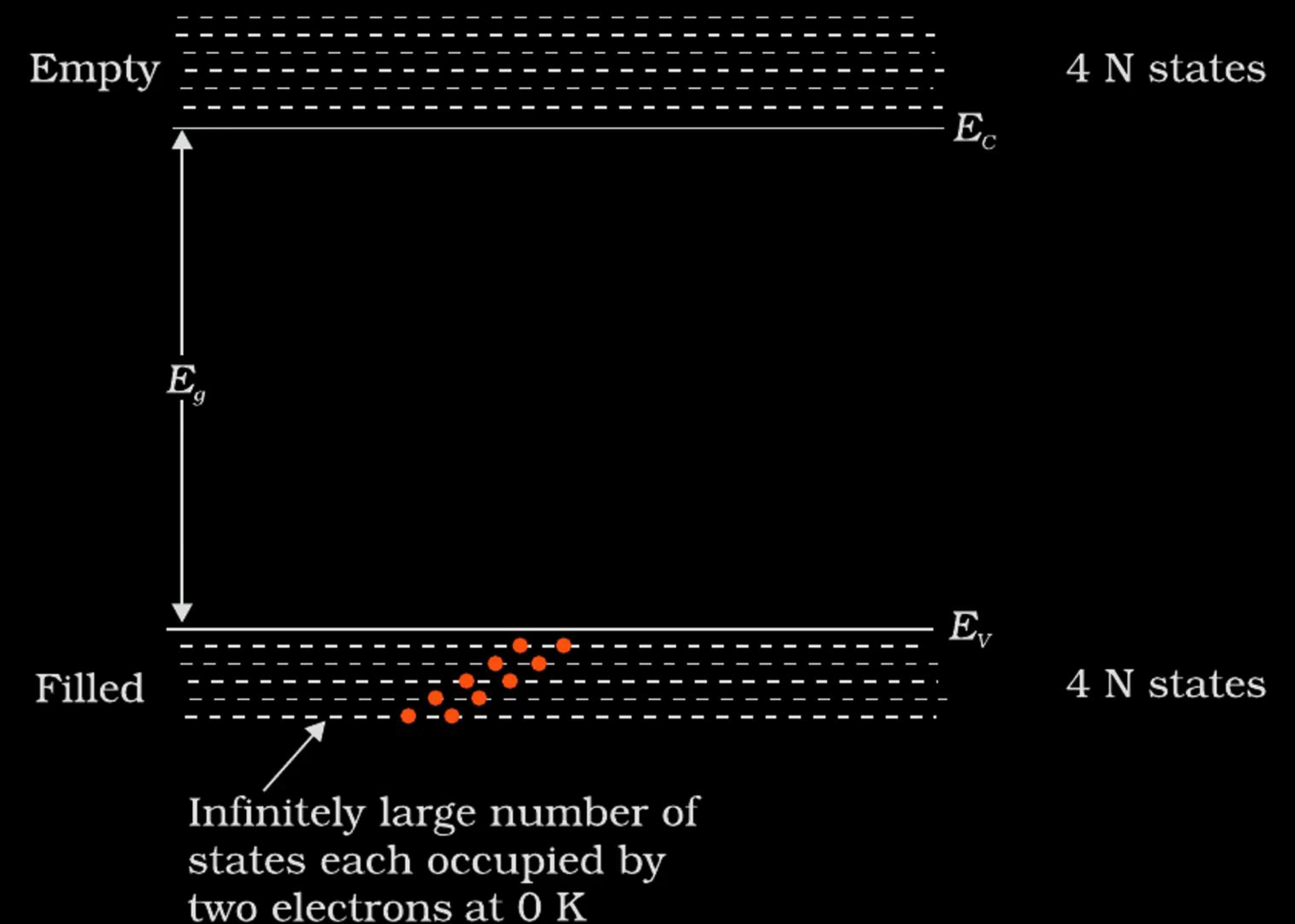
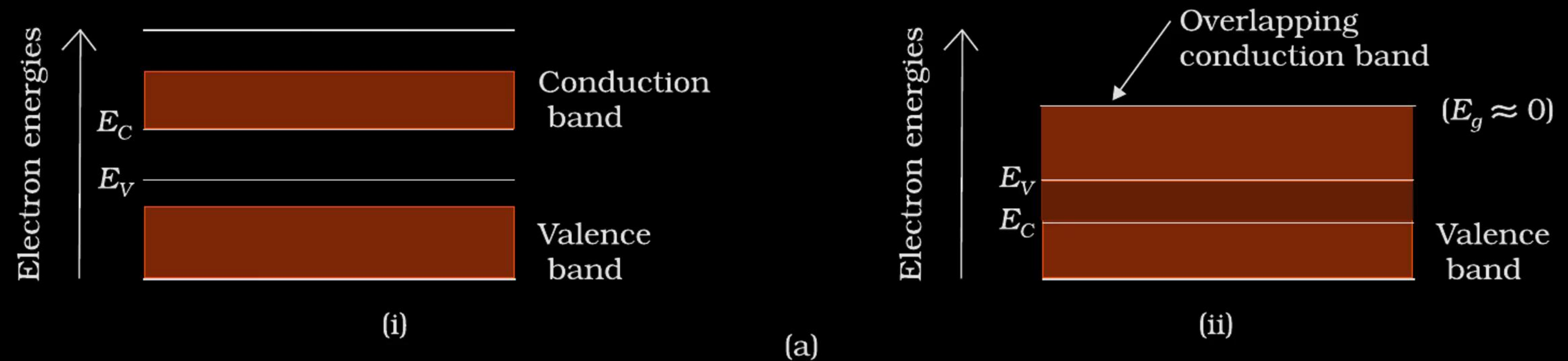


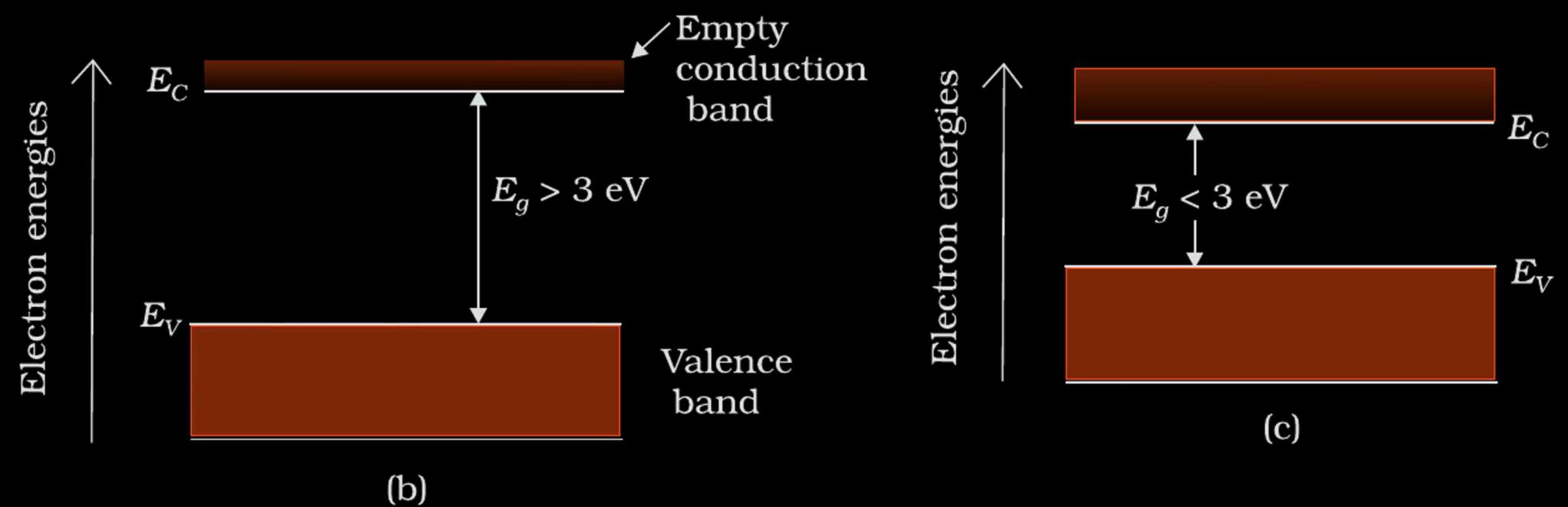
FIGURE 14.1 The energy band positions in a semiconductor at 0 K. The upper band, called the conduction band, consists of infinitely large number of closely spaced energy states. The lower band, called the valence band, consists of closely spaced completely filled energy states.

Classification on the Basis of Energy Bands

- **Metals:** CB and VB overlap or CB partially filled $\Rightarrow$ many free electrons

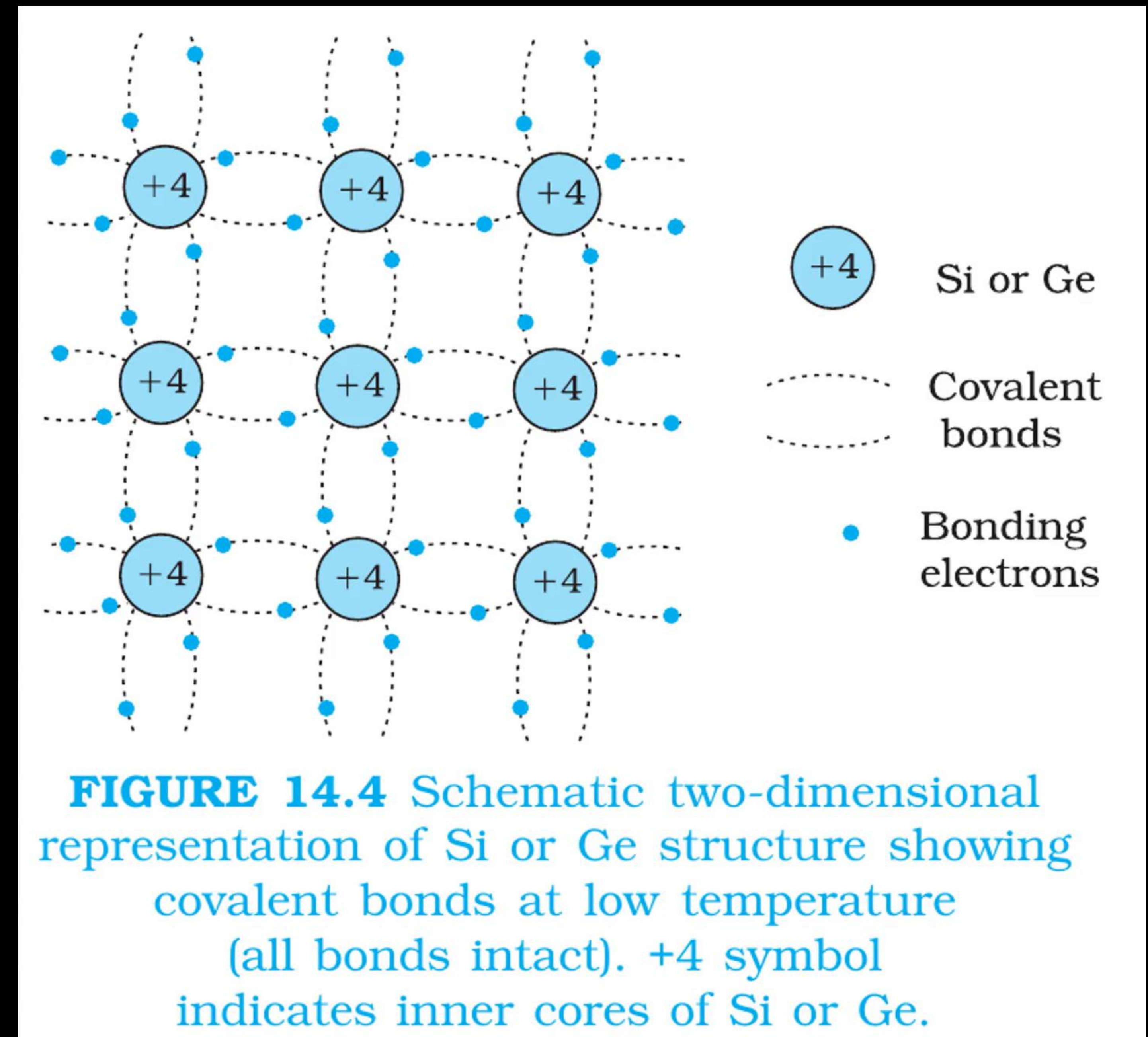


- **Insulators:** Large energy gap $E_g > 3 \text{ eV}$ between VB and CB $\Rightarrow$ no electron flow.
- **Semiconductors:** Small gap $E_g < 3 \text{ eV}$; some electrons can jump from VB to CB at room temperature.



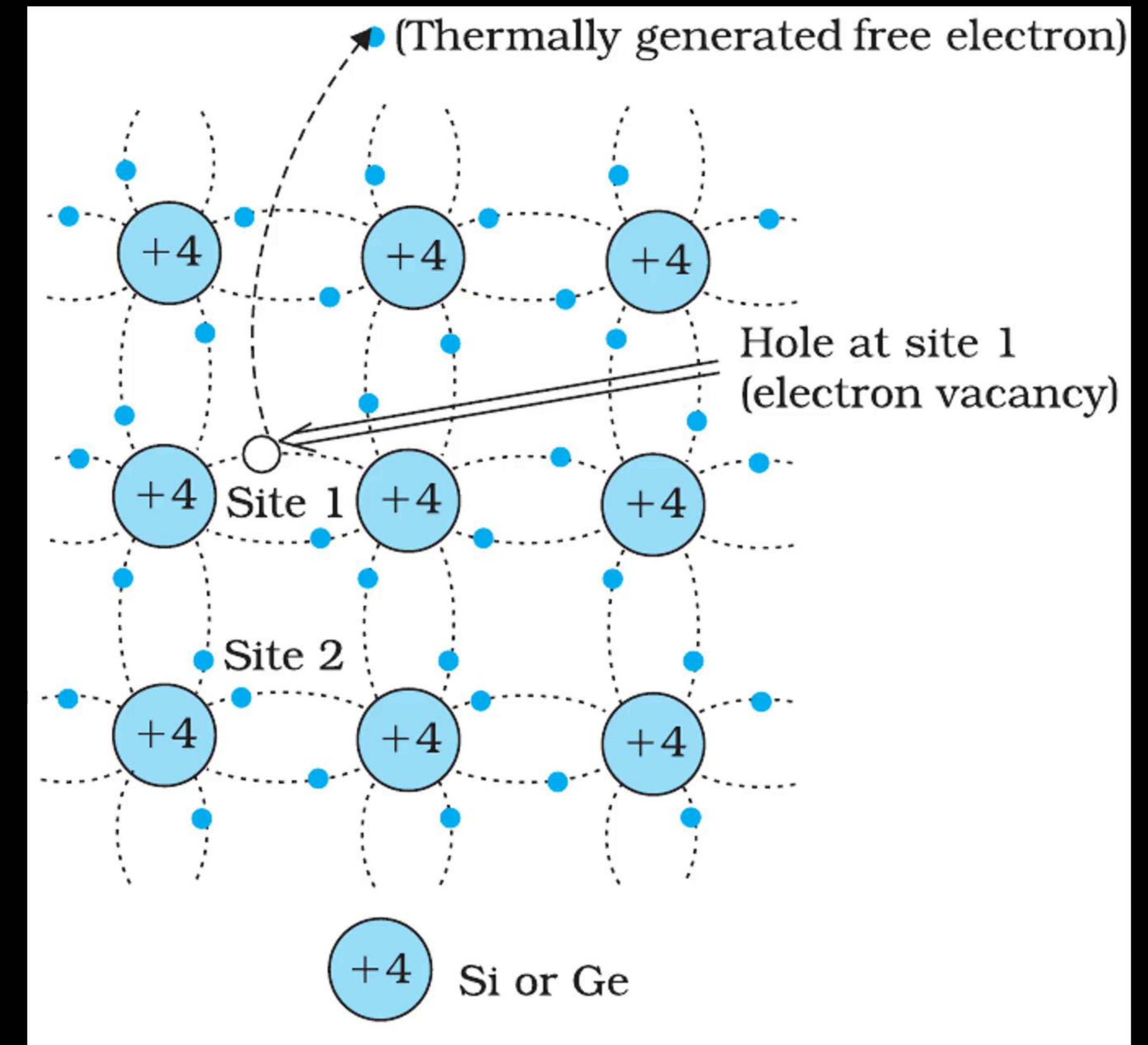
Intrinsic Semiconductor

- A pure semiconductor without any impurity is called an intrinsic semiconductor.
- Common examples: **Silicon (Si)** and **Germanium (Ge)**.
- Both Si and Ge have four valence electrons.
- In a crystal each atom shares its four valence electrons with four neighbouring atoms, forming covalent bonds.
- These make a stable lattice called the **diamond-like structure** (each atom is surrounded by four nearest neighbours).



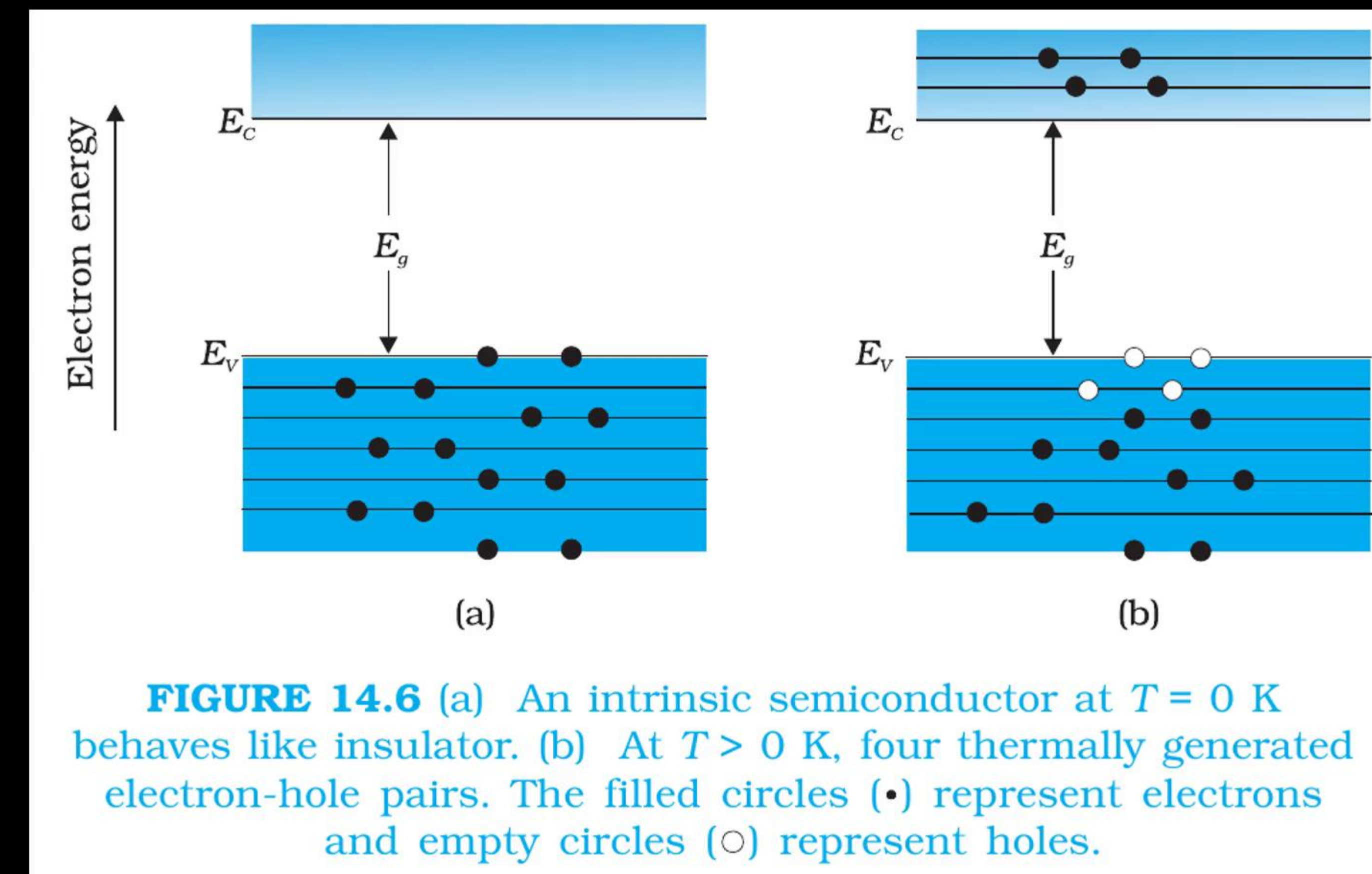
Intrinsic Semiconductor

- At 0 K: all valence electrons are tightly bound in covalent bonds; no free electrons are available for conduction.
- Thus at 0 K an intrinsic semiconductor behaves like an **insulator**.
- At room temperature thermal energy breaks some covalent bonds.
- When a bond breaks an electron becomes free (a conduction electron) and the missing electron leaves behind a vacancy called a **hole**.
- A **hole** behaves like an apparent particle with positive charge; holes and electrons both contribute to conduction.



Intrinsic Semiconductor

- Charge carriers: **Electrons** (-ve) in conduction band and **Holes** (+ve) in valence band.
- Current = electron current + hole current.
- At equilibrium the number of free electrons equals the number of holes: $n_e = n_h = n_i$, where n_i is intrinsic carrier concentration.
- Motion of holes: when an electron from a nearby bond fills a hole, the hole appears to move to the neighbour site; thus holes behave as if they move though actually they are vacancies.
- Recombination: free electron may fall into a hole restoring the bond. Generation and recombination occur continuously and at equilibrium their rates are equal.

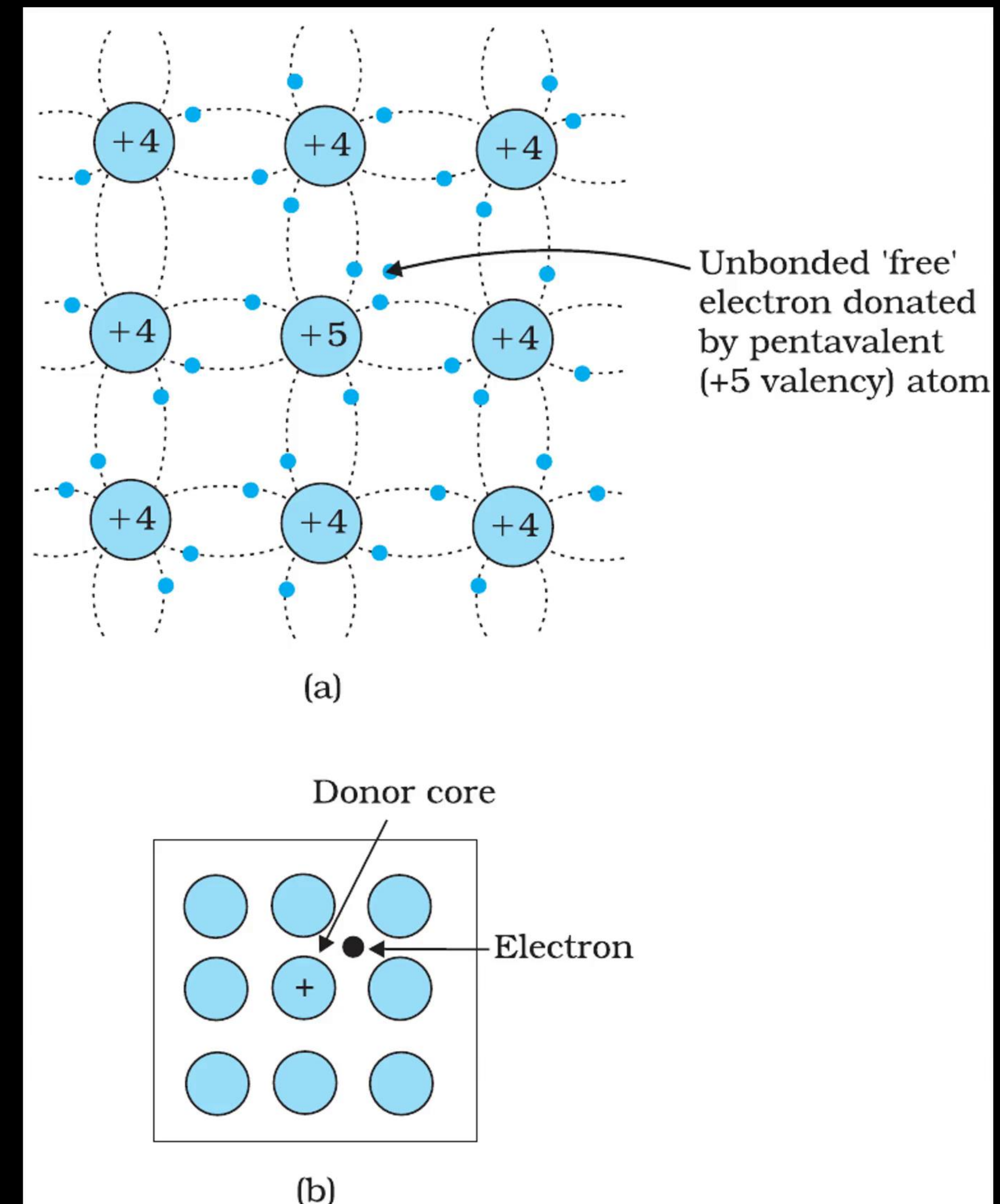


Extrinsic Semiconductor

- A pure (intrinsic) semiconductor has very low conductivity at room temperature.
- By adding a small amount of suitable impurity, conductivity is increased.
- Such a doped semiconductor is called an **extrinsic semiconductor**.
- The process of adding impurity is called **doping**.
- Typical doping: one impurity atom per 10^6 - 10^8 host atoms.
- The dopant has to be such that it does not distort the original pure semiconductor lattice.
- The sizes of the dopant and the semiconductor atoms should be nearly the same.

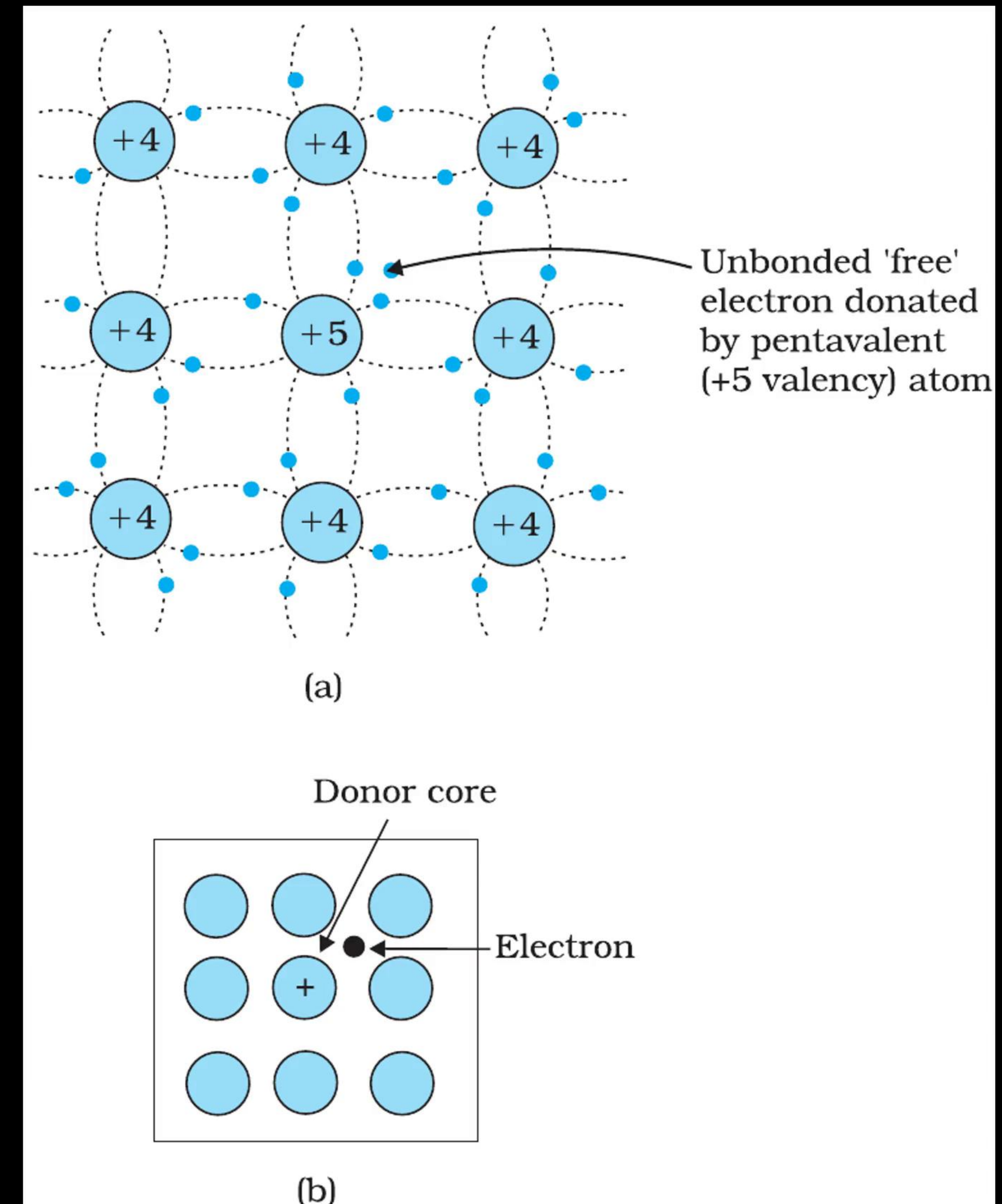
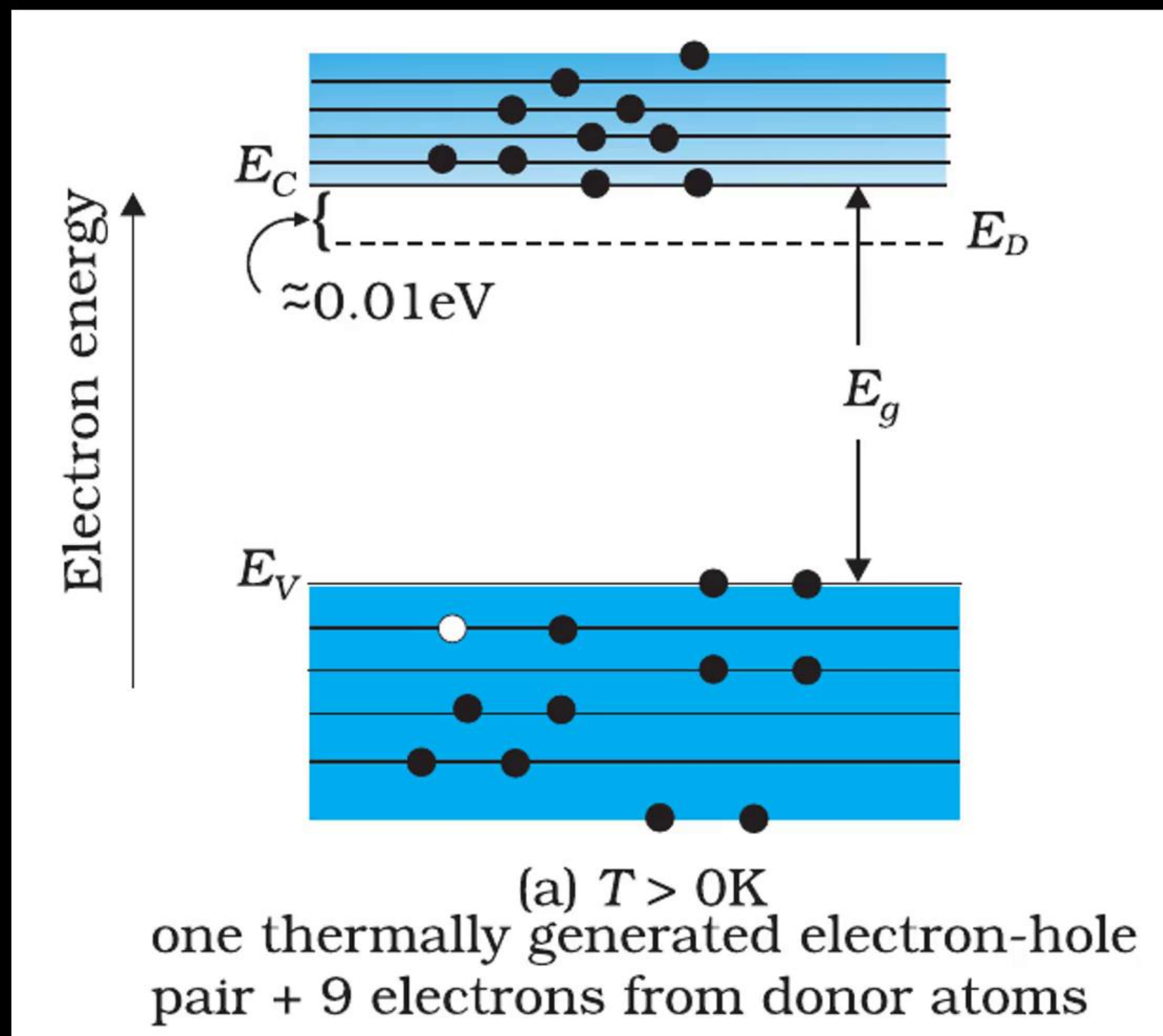
n-type Semiconductor

- Formed by doping with **pentavalent (valency 5) atoms** (P, As, Sb, Bi).
- Each has 5 valence electrons: 4 form covalent bonds, the 5th is extra and loosely bound.
- This extra electron is easily freed (energy required is ≈ 0.01 eV for Ge, and 0.05 eV for Si, to separate this electron from its atom) and contributes to conduction.
- The number of electrons made available for conduction by dopant atoms **depends strongly upon the doping level** and is independent of any **increase in ambient temperature**.



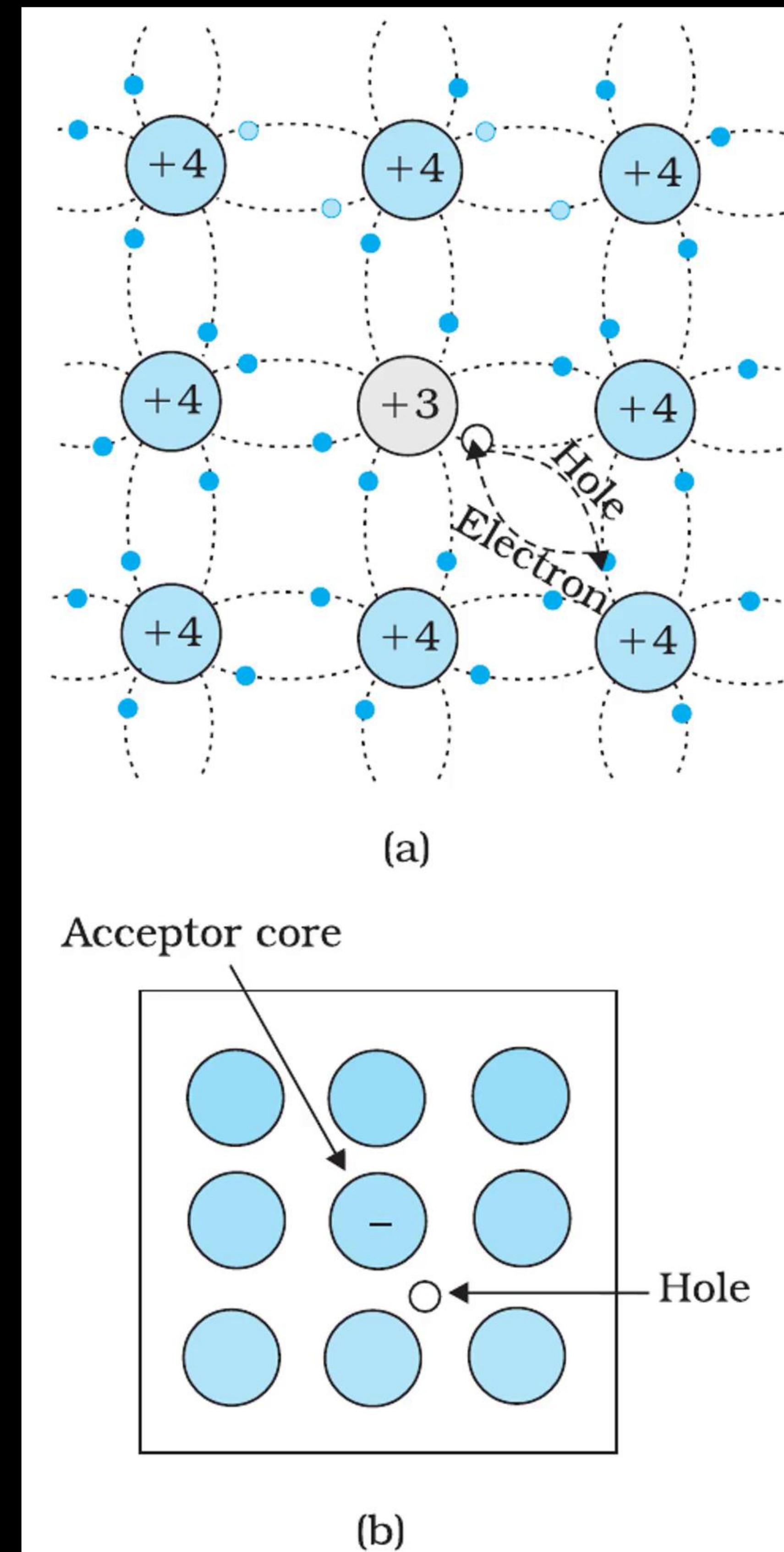
n-type Semiconductor

- **Electrons are majority carriers**, holes are minority carriers ($n_e \gg n_h$) and ($n_h < n_i$) due to increase in rate of recombination.
- **Donor energy level:** just below conduction band; requires very little energy to excite electron into CB.



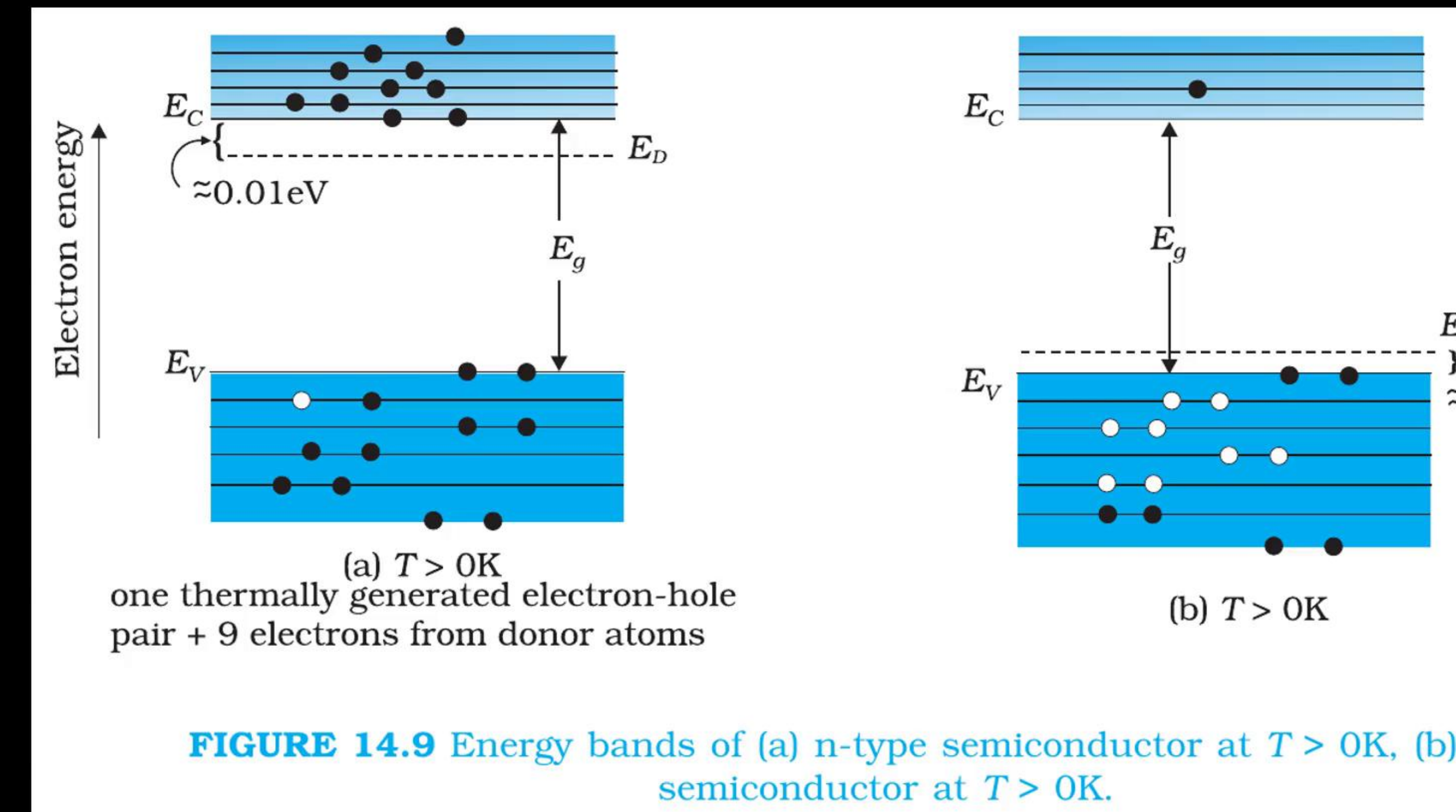
p-type Semiconductor

- Formed by doping with **trivalent (valency 3) atoms** (B, Al, Ga, In).
- Each has 3 valence electrons: form only 3 covalent bonds, leaving one incomplete bond.
- This incomplete bond behaves like a **hole** (positive charge carrier).
- **Holes are majority carriers**, electrons are minority carriers. ($n_h \gg n_e$)
- **Acceptor energy level:** just above valence band; a small energy excites electron into this level, creating a hole in VB.



Modification in Energy Bands & Carrier Relation

- Doping introduces **new energy levels** in the band gap.
- In n-type: donor level lies just below conduction band.
- In p-type: acceptor level lies just above valence band.
- Thus electrons can easily be excited to CB (n-type) or holes easily created in VB (p-type).
- **Mass-action law:** $n_e \cdot n_h = n_i^2$, valid at a given temperature.
- n-type: $n_e \gg n_h$, but $n_e n_h = n_i^2$; p-type: $n_h \gg n_e$, but $n_e n_h = n_i^2$.



Example 14.2 : Suppose a pure Si crystal has 5×10^{28} atoms m^{-3} . It is doped by 1 ppm concentration of pentavalent As. Calculate the number of electrons and holes. Given that $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$.

Solution :

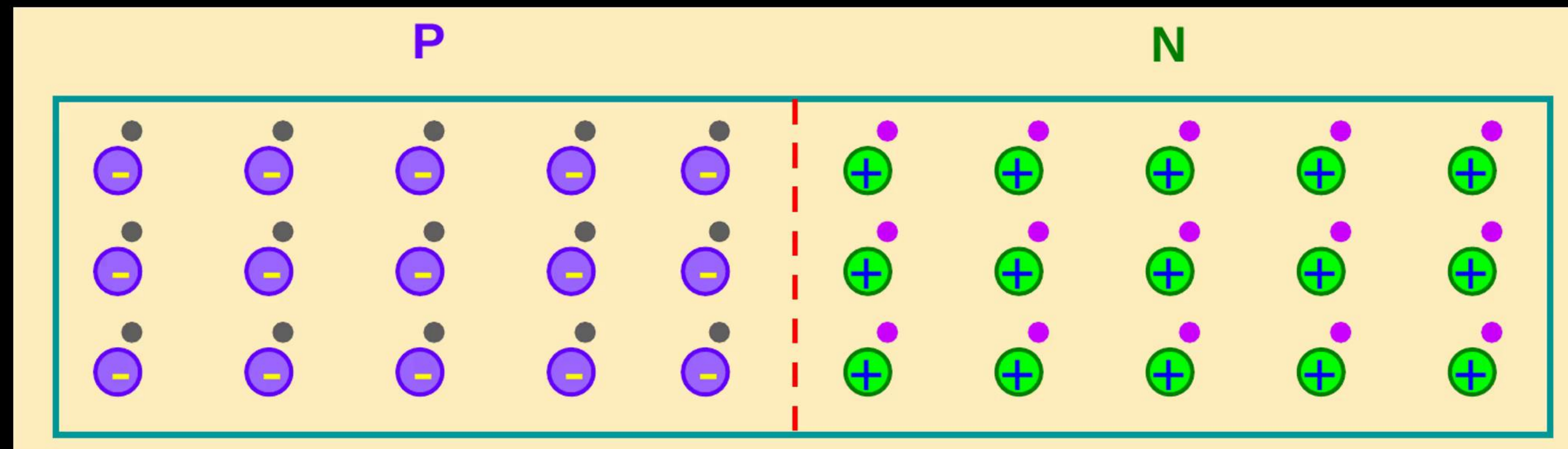
- Note that thermally generated electrons ($n_i \approx 10^{16} \text{ m}^{-3}$) are negligibly small as compared to those produced by doping.
- Therefore, $n_e \approx N_D$.
- Since $n_e n_h = n_i^2$, The number of holes

$$n_h = \frac{2.25 \times 10^{32}}{5 \times 10^{22}}$$

$$n_h \approx 4.5 \times 10^9 \text{ m}^{-3}$$

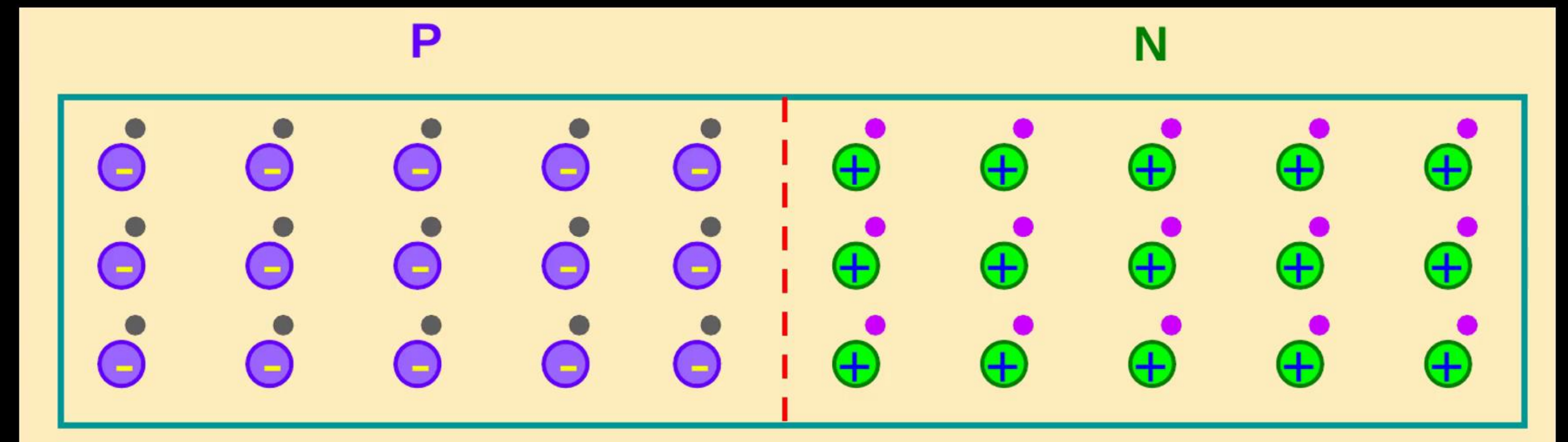
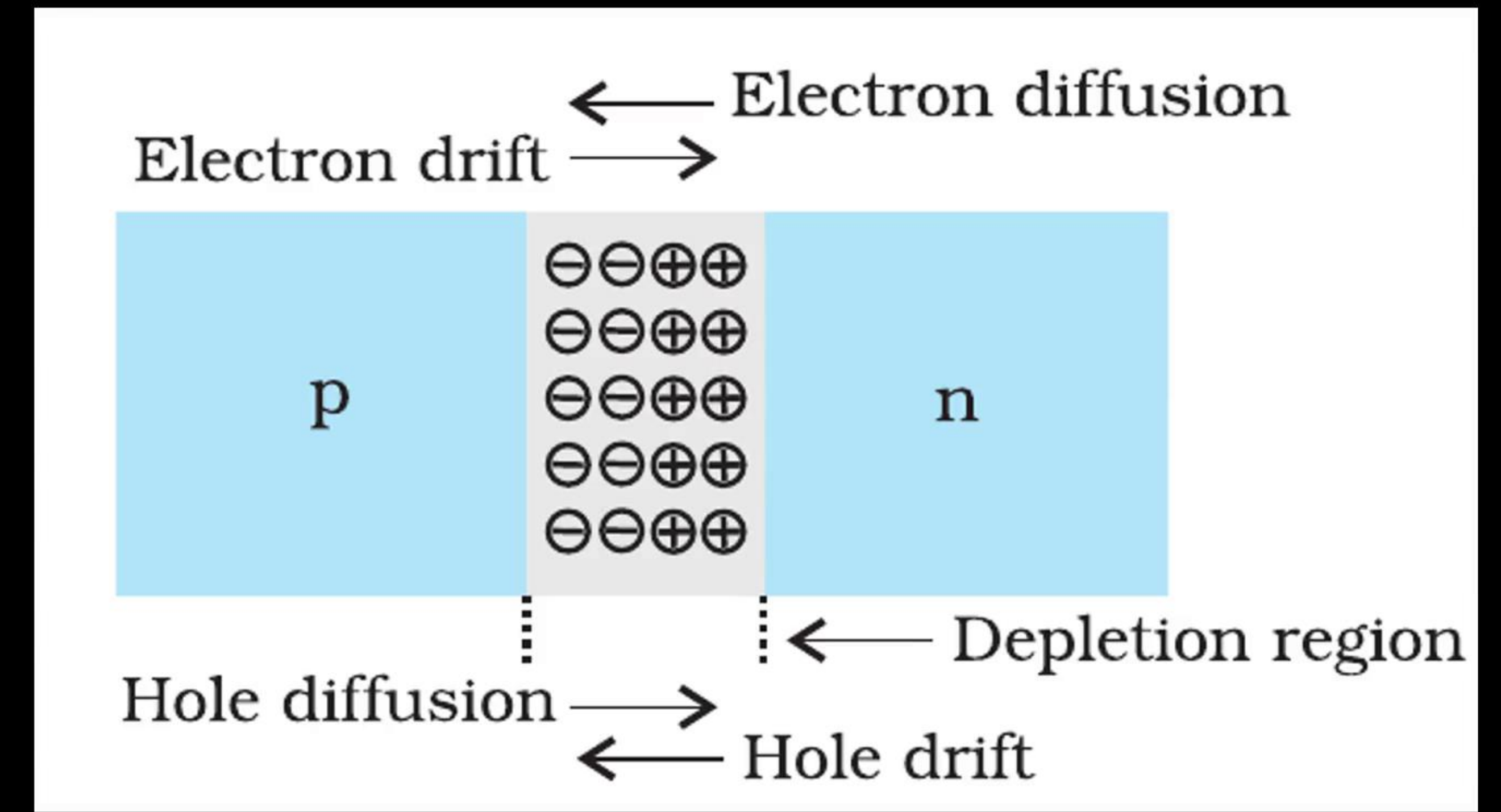
p-n Junction: Formation

- A **p-n junction** is formed when a p-type semiconductor is joined with an n-type semiconductor.
- It is the basic building block of semiconductor devices like diodes and transistors.
- In p-type: **holes are majority carriers**, electrons minority.
- In n-type: **electrons are majority carriers**, holes minority.
- When p-type and n-type are joined, electrons diffuse from n to p and holes from p to n due to concentration gradient.



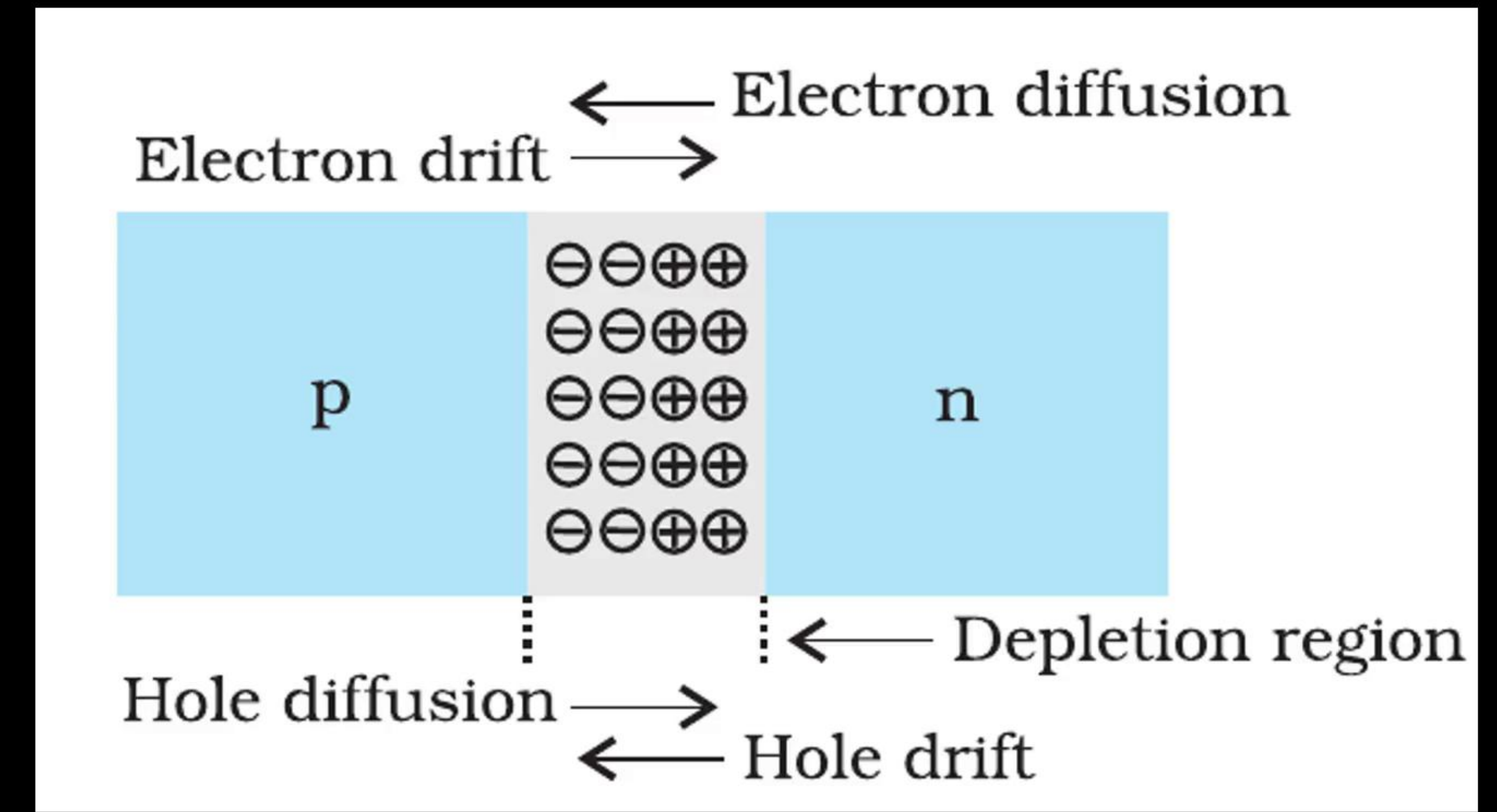
Diffusion in p-n Junction

- In n-type: electron concentration is high;
in p-type: hole concentration is high.
- Due to concentration gradient, electrons diffuse from n-side to p-side.
- Similarly, holes diffuse from p-side to n-side.
- This movement of majority carriers across the junction is called the **diffusion current**.
- Diffusion is the process by which majority carriers (electrons from n-side and holes from p-side) move across the p-n junction due to concentration gradient, producing a diffusion current.



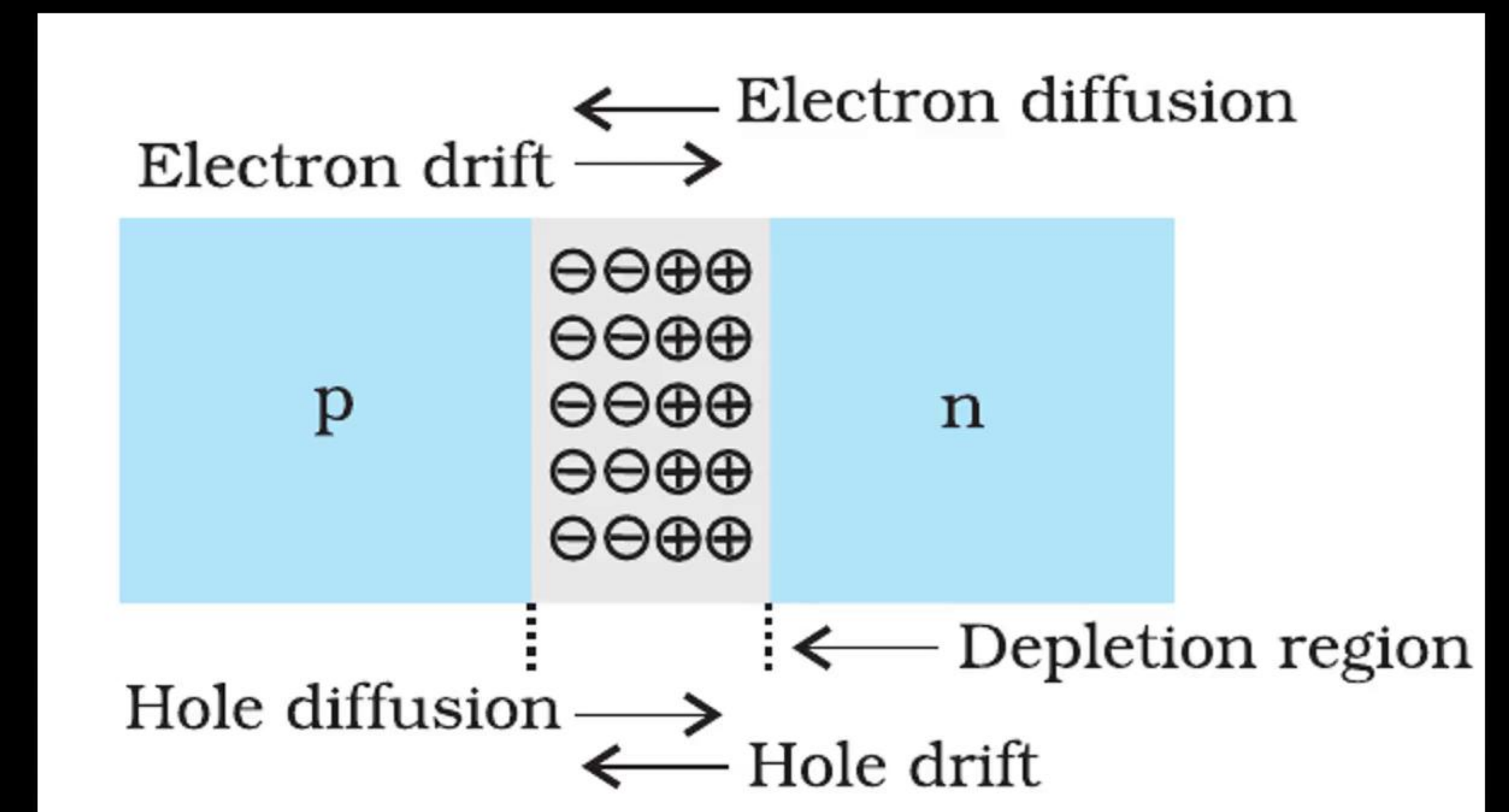
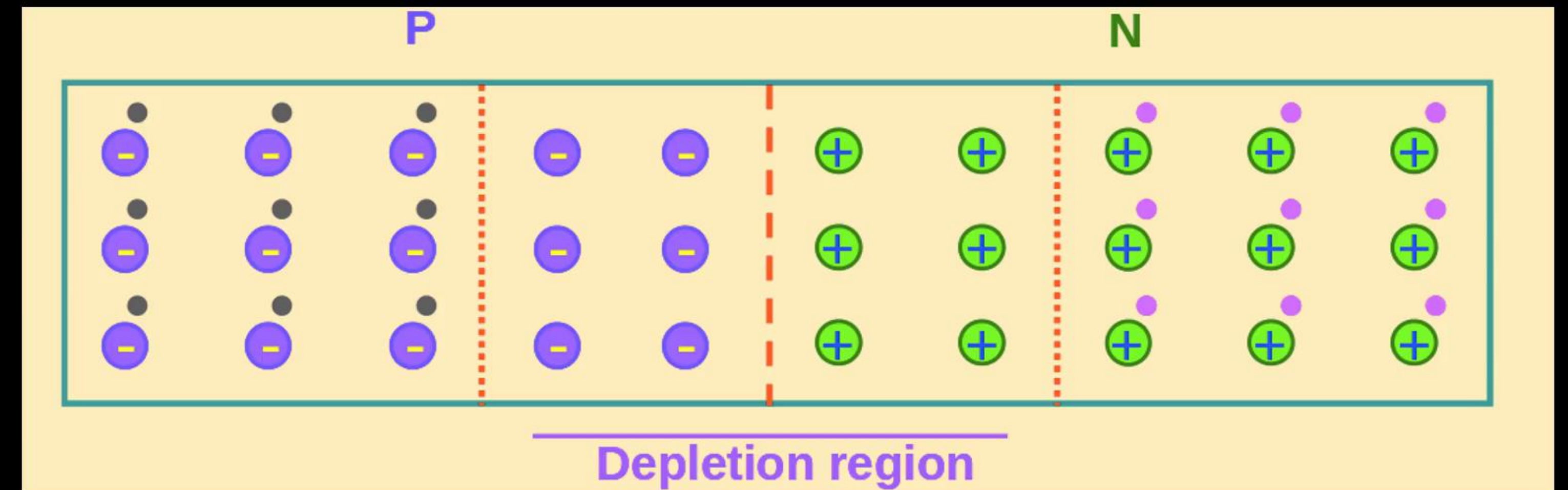
Drift in p-n Junction

- As diffusion occurs, electrons leaving n-side leave behind immobile **positive donor ions**.
- Holes leaving p-side leave behind immobile **negative acceptor ions**.
- These fixed ions create an **electric field** directed from n-side to p-side.
- This electric field pushes electrons (minority carriers) from p-side, back to n-side and holes from n-side, back to p-side.
- The resulting current opposite to diffusion current is called the **drift current**.
- Drift is the process by which minority carriers move across the p-n junction under the influence of the electric field in the depletion region, producing a drift current opposite to the diffusion current.



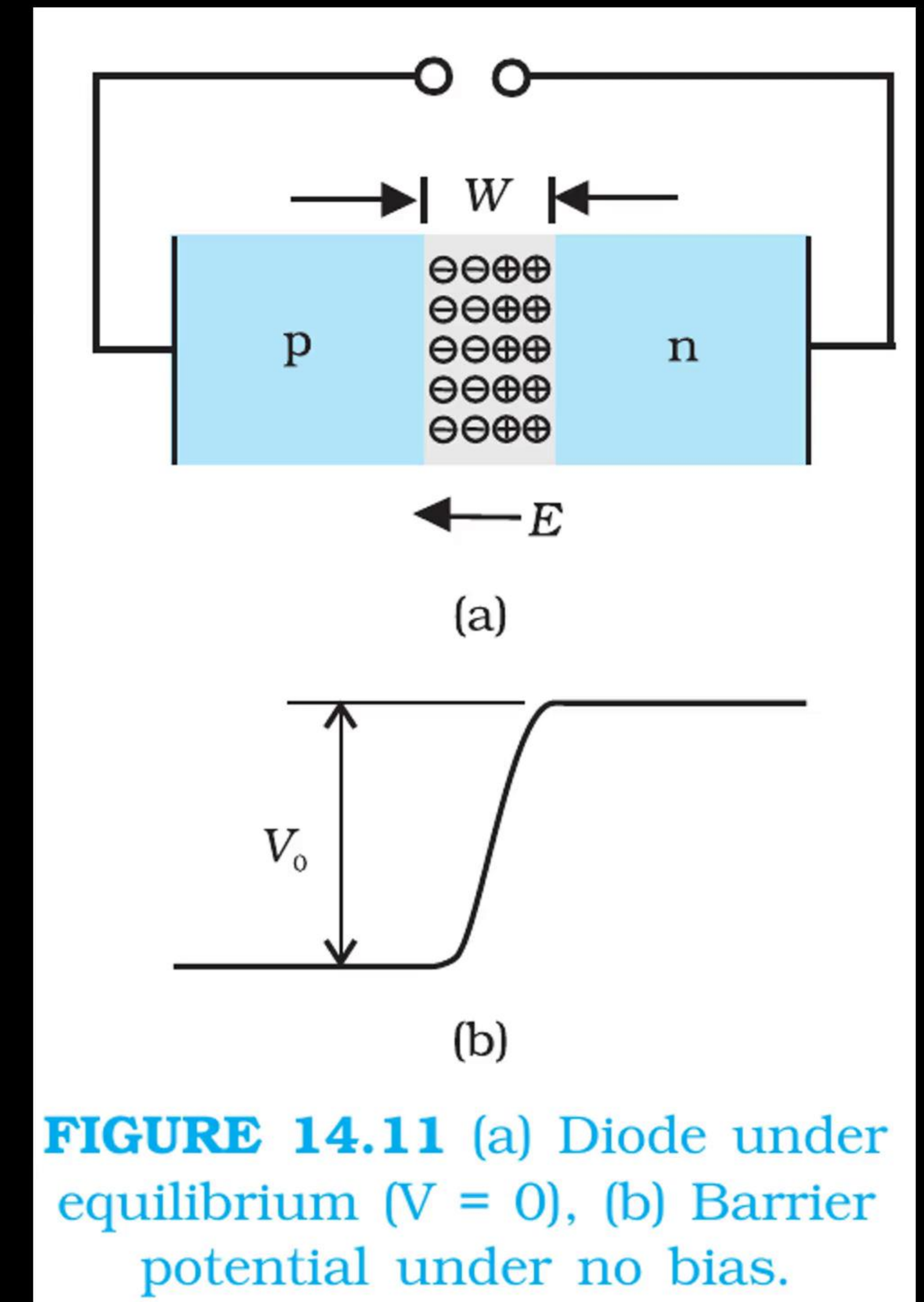
Depletion Region

- Near the junction, mobile charge carriers recombine and the region becomes depleted.
- This region contains only immobile ionised donor and acceptor ions.
- The zone is called the **depletion region** or space-charge region.
- No free charge carriers exist inside this region.
- The depletion region acts as a barrier to further carrier diffusion.
- The depletion region is the space-charge region near the p-n junction that is depleted of mobile charge carriers but contains immobile ions, thus preventing further diffusion.



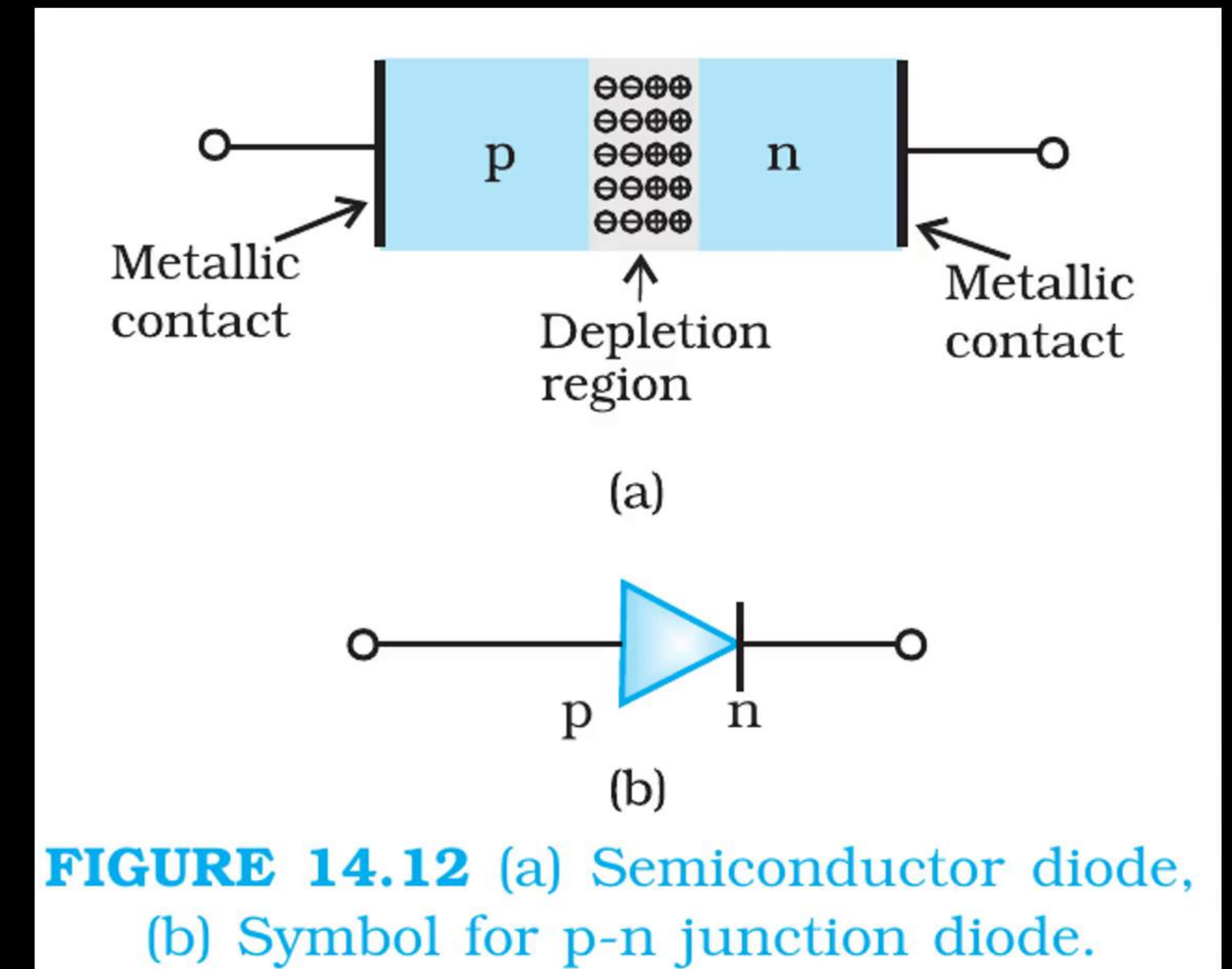
Barrier Potential

- The electric field across the depletion region creates a **potential barrier**.
- This barrier opposes further movement of majority carriers across the junction.
- Typical barrier potential: $\sim 0.7\text{ V}$ for Si, $\sim 0.3\text{ V}$ for Ge.
- At equilibrium: diffusion current equals drift current.
- Thus the net current across the junction is zero.
- The barrier potential is the potential difference developed across the depletion region of a p-n junction that prevents further movement of majority carriers.



Semiconductor Diode

- A **semiconductor diode** is a p-n junction with two terminals.
- p-side is called **anode**, n-side is called **cathode**.
- Current flows easily in one direction (forward bias) but not in the other (reverse bias).
- Thus, the diode acts as a **rectifier** (converts AC to DC).



Forward Bias

- In forward bias: p-side connected to +ve, n-side to -ve terminal of battery.
- External bias opposes built-in potential barrier.
- Depletion region width decreases and barrier height reduces to $(V_0 - V)$.
- **Minority carrier injection:** electrons from n-side cross into p-side, holes from p-side cross into n-side.
- Negligible current until applied voltage $\approx$ barrier potential (0.7 V for Si, 0.3 V for Ge).
- Beyond threshold, large current flows $\rightarrow$ diode becomes conducting.

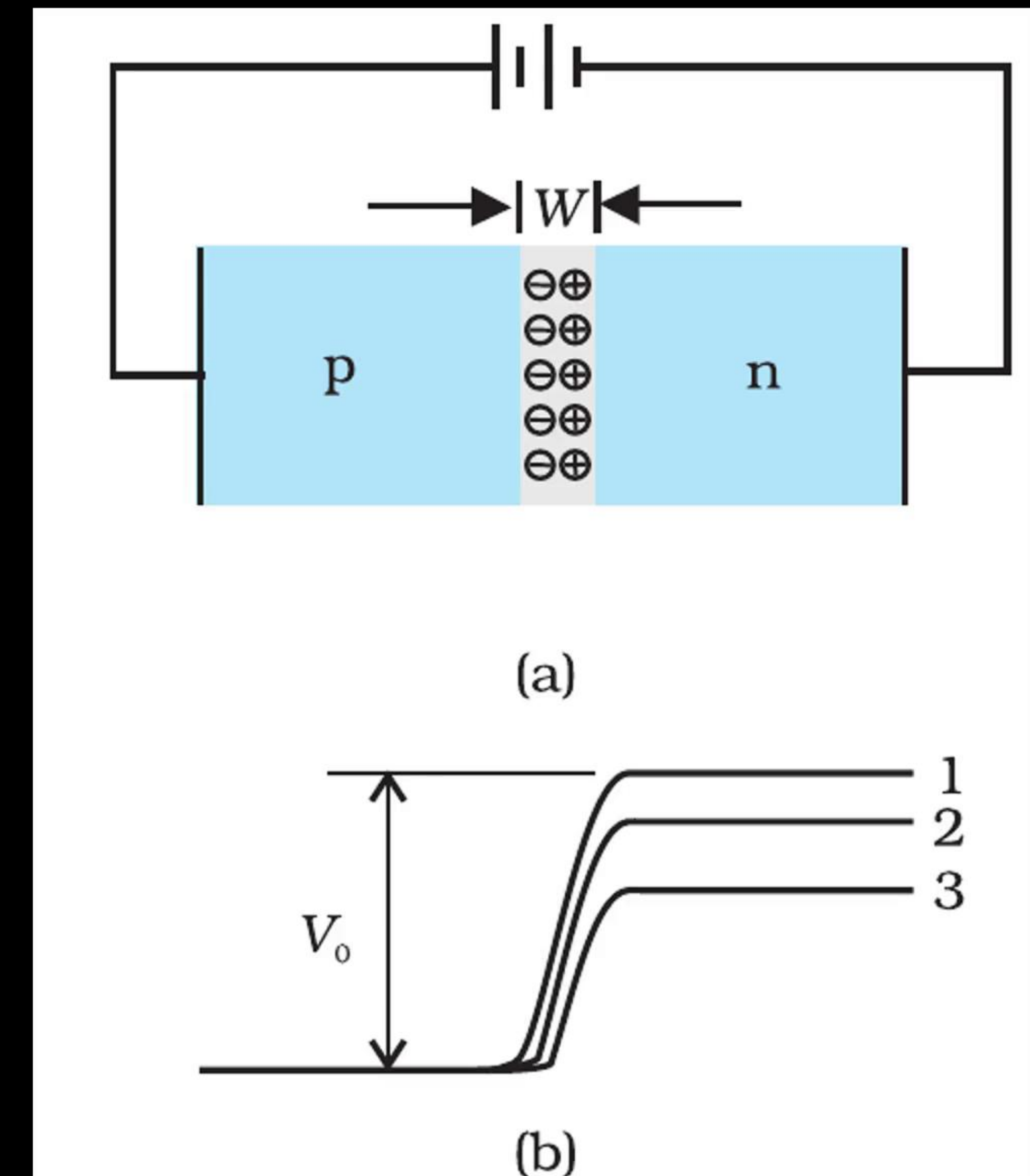


FIGURE 14.13 (a) p-n junction diode under forward bias, (b) Barrier potential (1) without battery, (2) Low battery voltage, and (3) High voltage battery.

Reverse Bias

- In reverse bias: p-side connected to $-ve$, n-side to $+ve$ terminal of battery.
- External bias aids built-in potential barrier.
- Depletion region width increases and barrier height increases ($V_0 + V$).
- Majority carriers cannot cross the junction.
- Only a very small reverse saturation current flows due to minority carriers.
- At high reverse voltage, diode undergoes **break-down** (sudden large current).

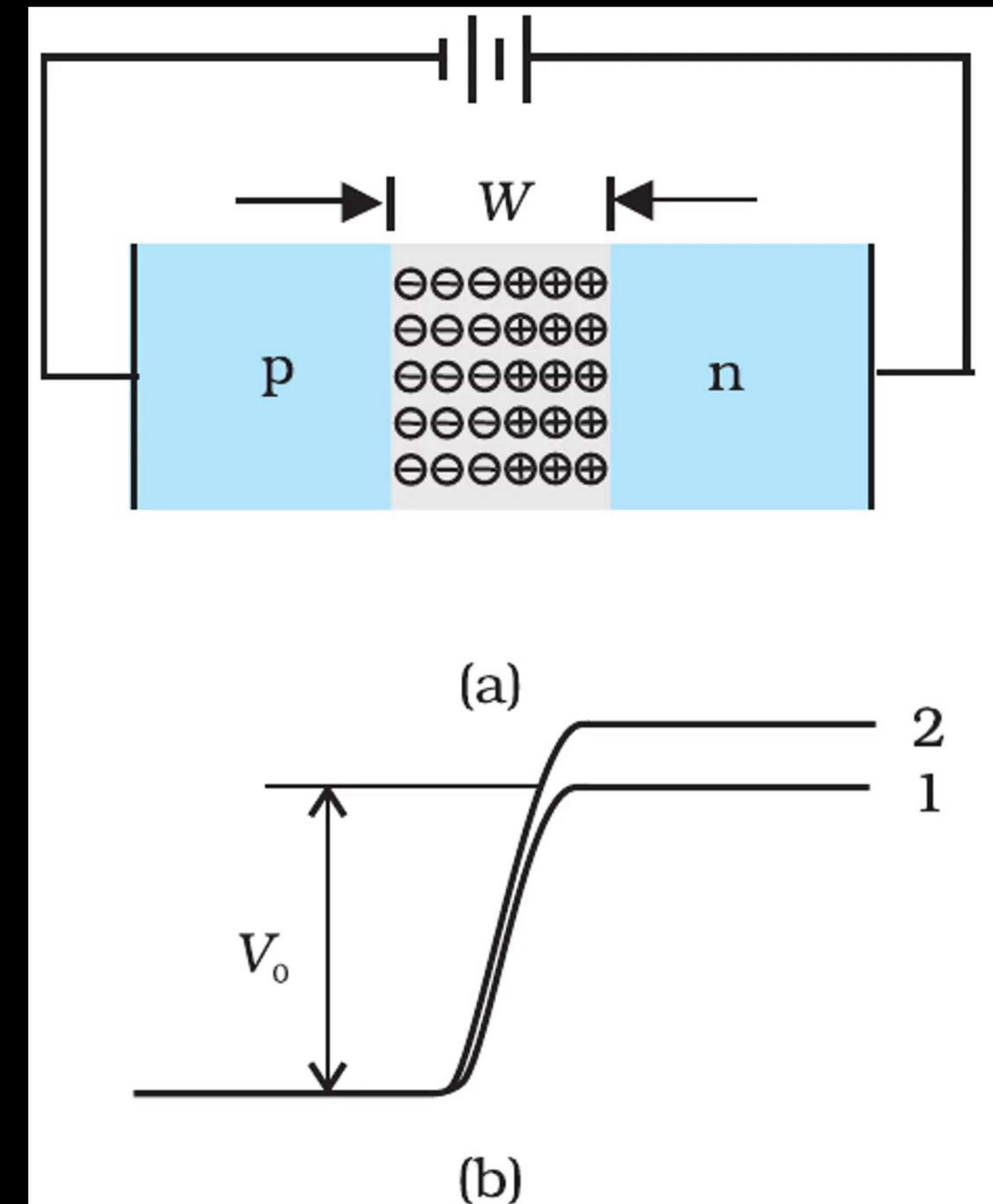


FIGURE 14.15 (a) Diode under reverse bias, (b) Barrier potential under reverse bias.

V-I Characteristics of Diode

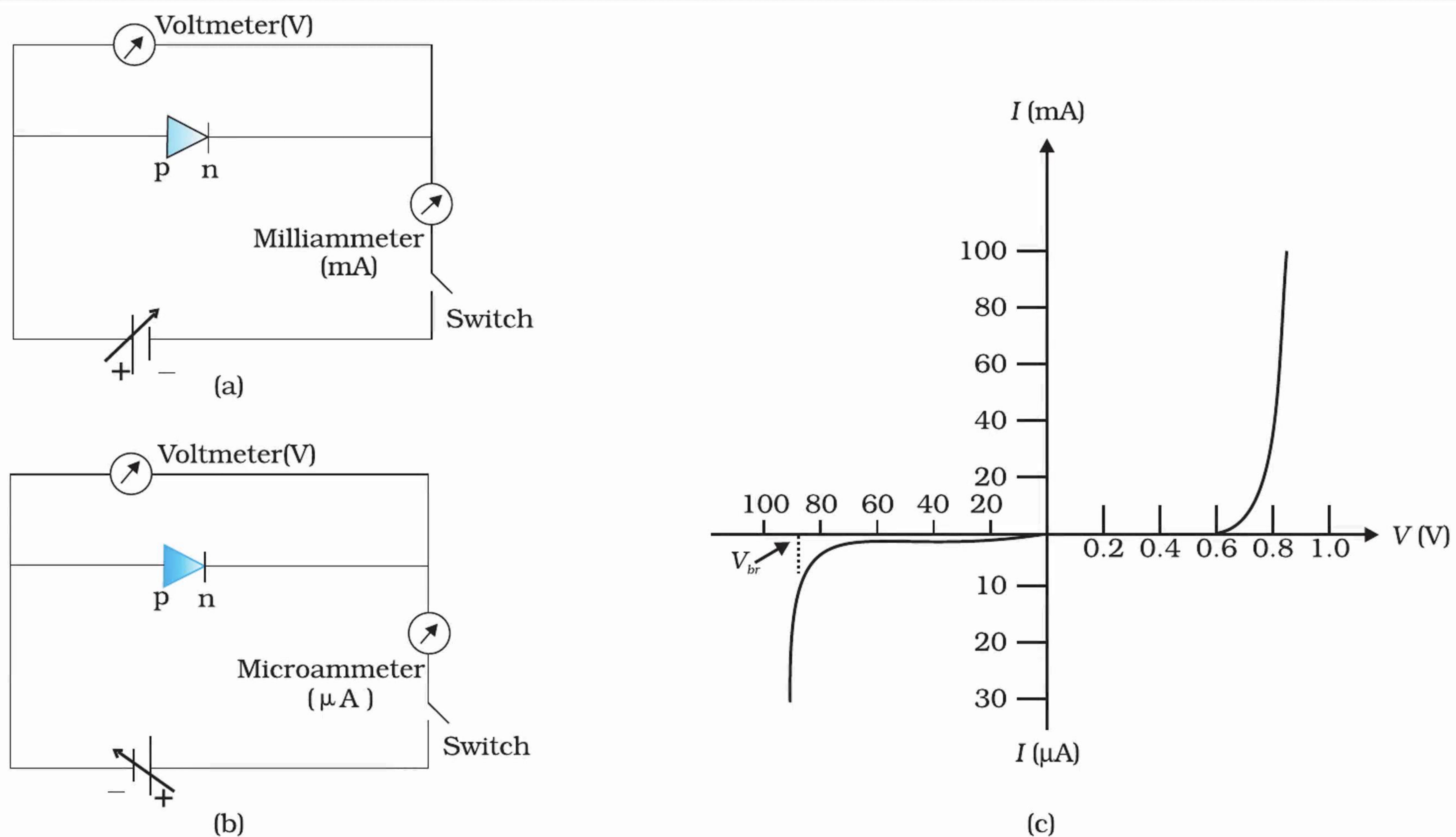
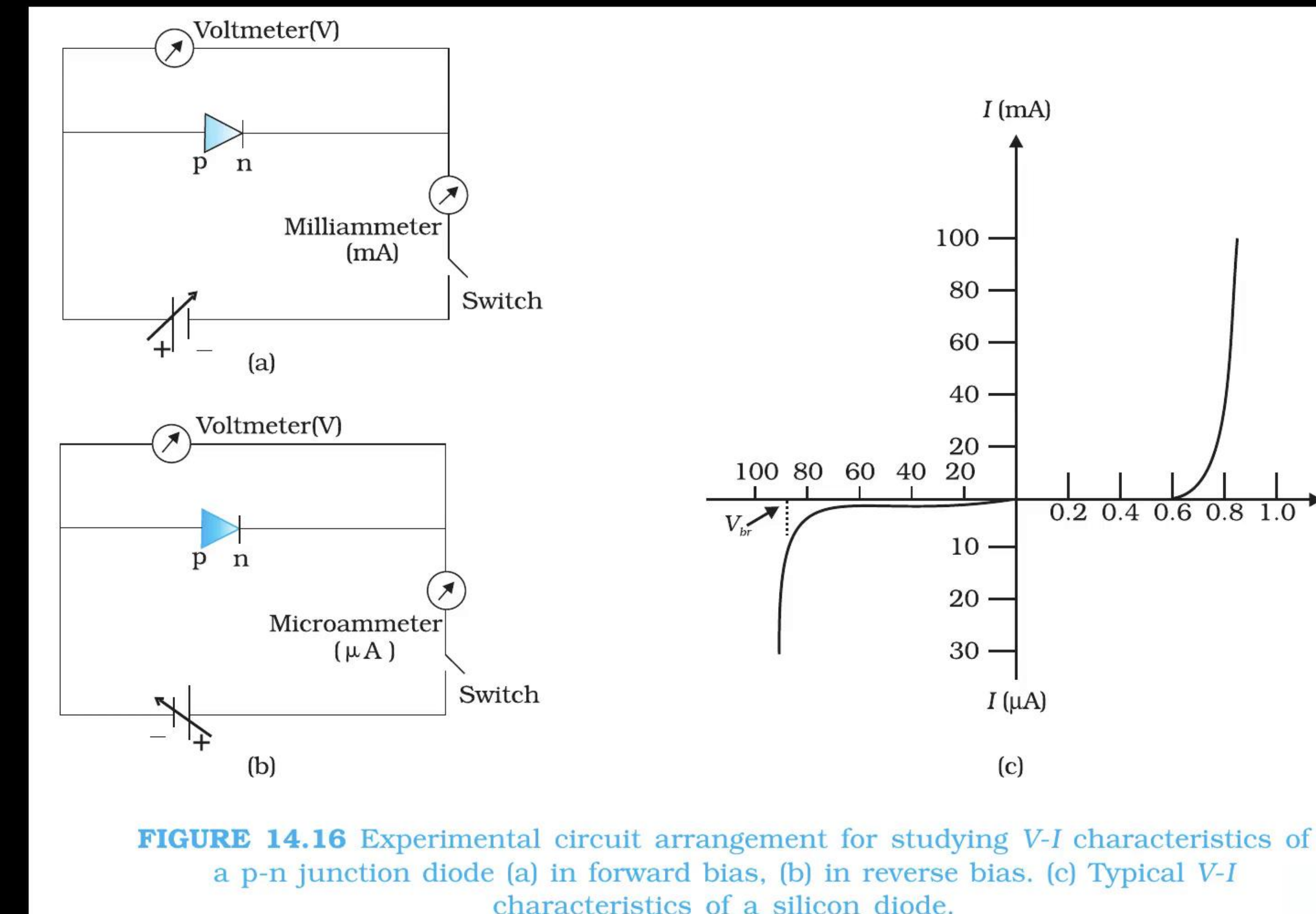


FIGURE 14.16 Experimental circuit arrangement for studying V-I characteristics of a p-n junction diode (a) in forward bias, (b) in reverse bias. (c) Typical V-I characteristics of a silicon diode.

V-I Characteristics of Diode

- **Forward Bias:** Current negligible until applied voltage $\approx$ barrier potential.
- Beyond threshold (0.7 V Si, 0.3 V Ge), current (in mA) rises steeply with voltage.
- **Reverse Bias:** Only small reverse saturation current (few μA) flows due to minority carriers.
- The general purpose diode are not used beyond the reverse saturation current region.
- At high reverse voltage, breakdown occurs: sharp rise in current.
- **Dynamic resistance** : The ratio of small change in voltage ΔV to a small change in current ΔI : $r_d = \frac{\Delta V}{\Delta I}$



Example 14.4 The V - I characteristic of a silicon diode is shown in the Fig. 14.17. Calculate the resistance of the diode at (a) $I_D = 15$ mA and (b) $V_D = -10$ V.

Solution Considering the diode characteristics as a straight line between $I = 10$ mA to $I = 20$ mA passing through the origin, we can calculate the resistance using Ohm's law.

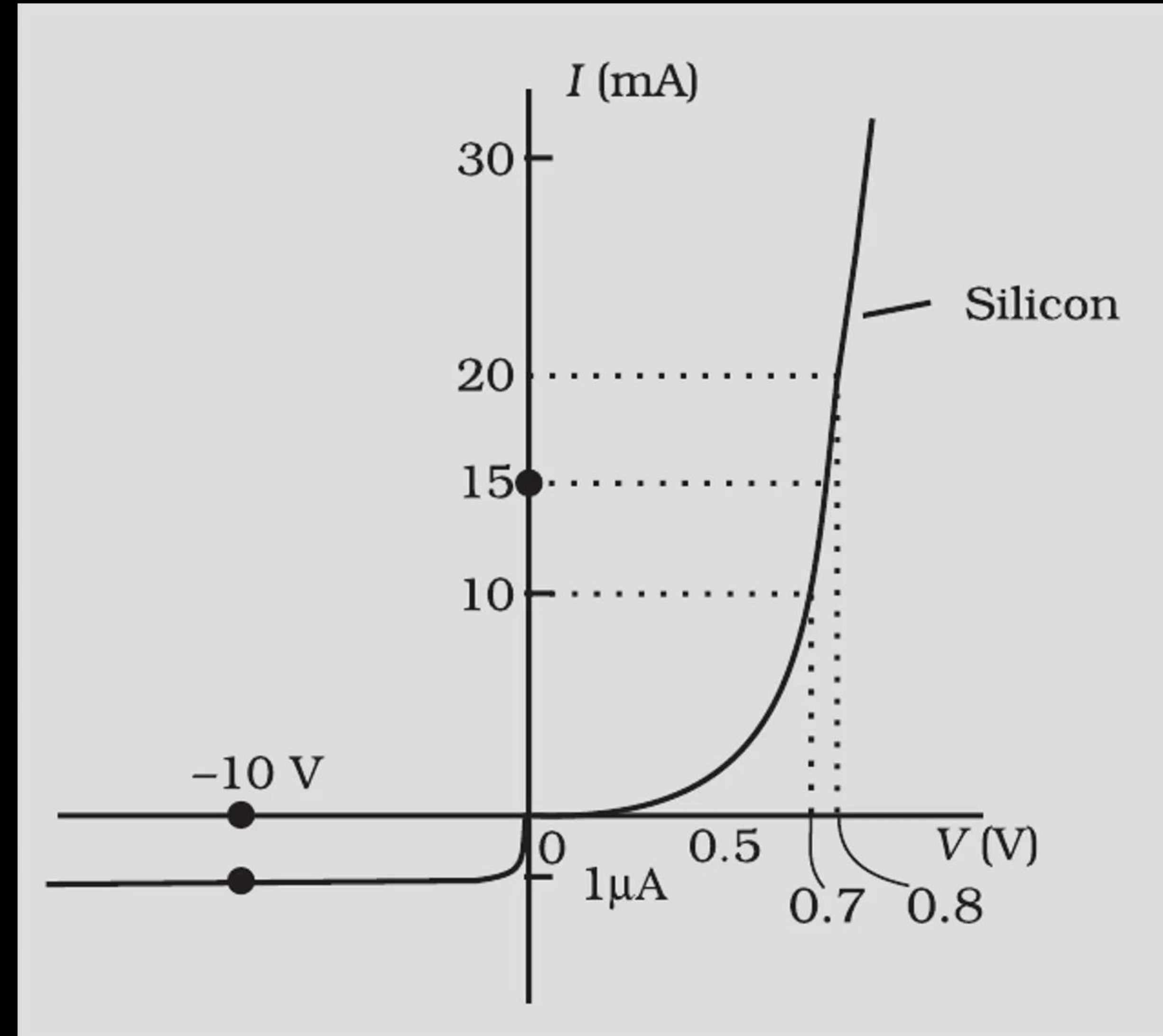
(a) From the curve, at $I = 20$ mA, $V = 0.8$ V; $I = 10$ mA, $V = 0.7$ V

$$r_{fb} = \Delta V / \Delta I = 0.1 \text{ V} / 10 \text{ mA} = 10 \text{ } \Omega$$

(b) From the curve at $V = -10$ V, $I = -1$ μ A,

Therefore,

$$r_{rb} = 10 \text{ V} / 1 \mu\text{A} = 1.0 \times 10^7 \text{ } \Omega$$



Application of Junction Diode as a Rectifier

- A **rectifier** is a device which converts alternating current (AC) into direct current (DC).
- It works on the property of a **p-n junction diode**:
 - Forward bias $\rightarrow$ diode conducts current (low resistance).
 - Reverse bias $\rightarrow$ diode blocks current (high resistance).
- Thus, current flows only in one direction $\rightarrow$ diode acts as a rectifier.

Half-Wave Rectifier

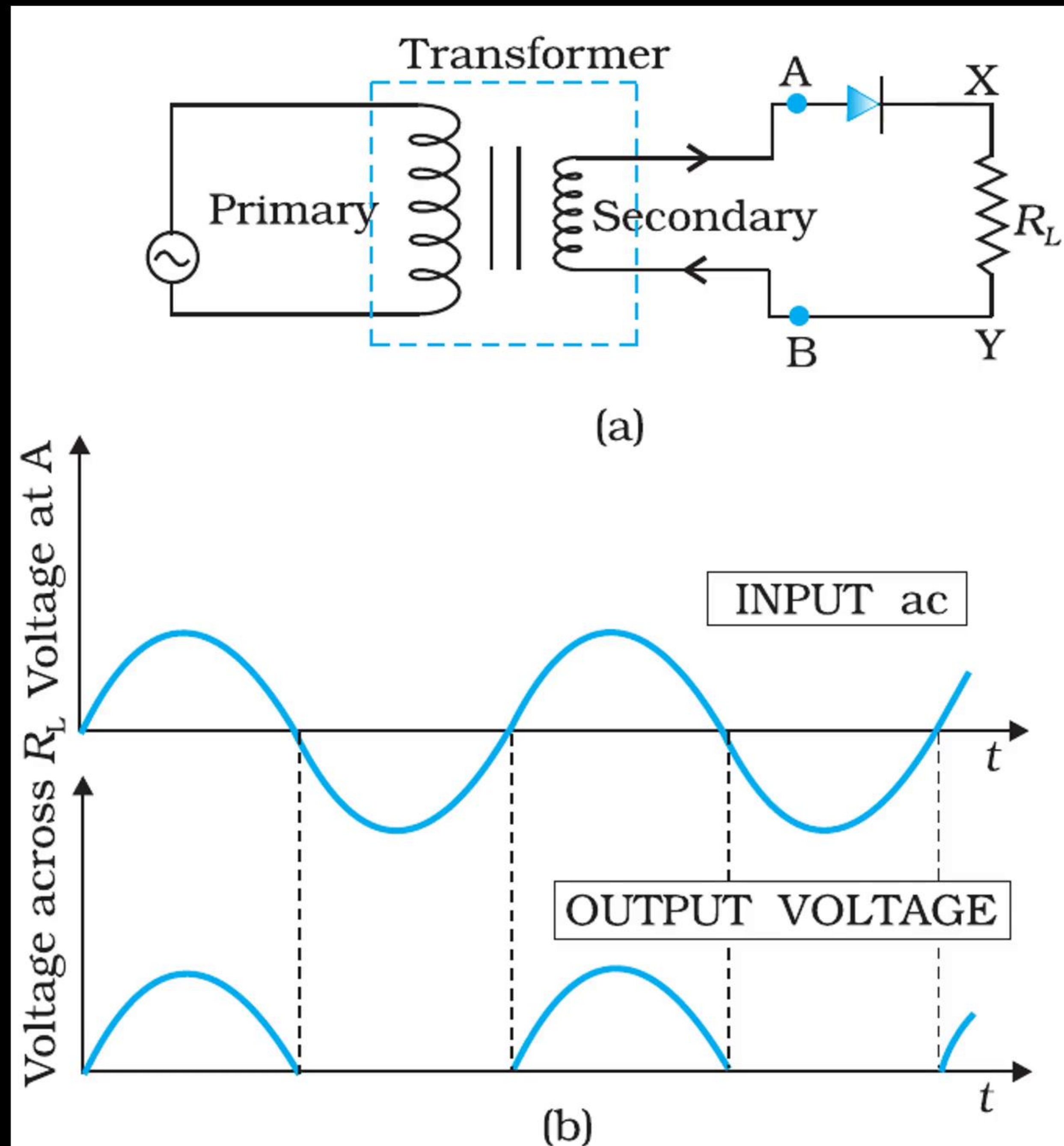


FIGURE 14.18 (a) Half-wave rectifier circuit, (b) Input ac voltage and output voltage waveforms from the rectifier circuit.

- A half-wave rectifier is a type of rectifier that **allows only one-half cycle** of an AC voltage waveform to pass while **blocking the other half cycle**.
- **Circuit:** Diode in series with load resistor R_L , input through transformer.
- **Working:**
 - Positive half-cycle → diode forward biased → current through R_L .
 - Negative half-cycle → diode reverse biased → no current flows.
- **Output:** Only positive half-cycles appear → **pulsating DC**.

Half-Wave Rectifier: Waveforms

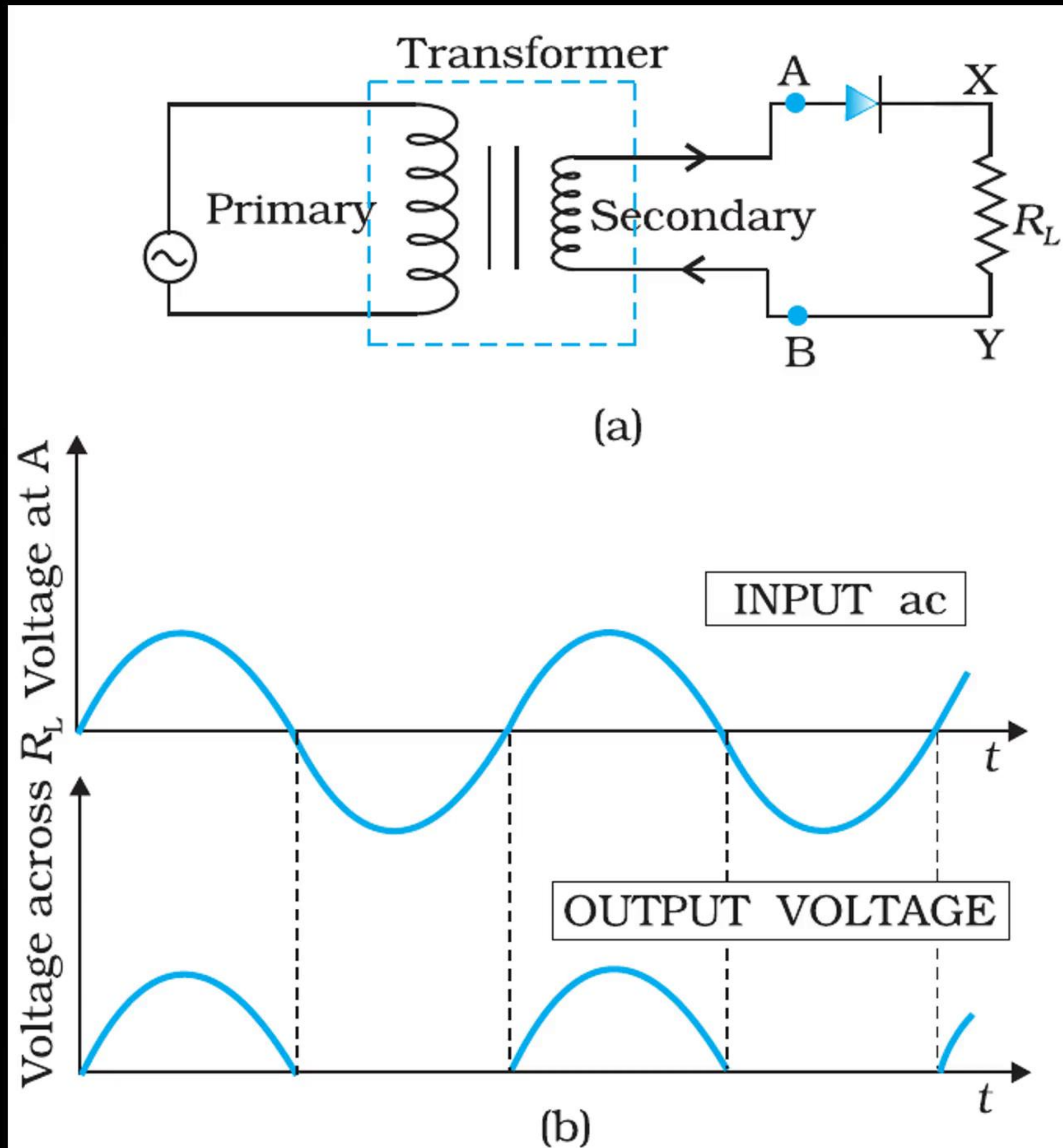


FIGURE 14.18 (a) Half-wave rectifier circuit, (b) Input ac voltage and output voltage waveforms from the rectifier circuit.

- **Input waveform:** Complete sinusoidal AC.
- **Output waveform:** Only positive half-cycles across R_L .
- Thus, DC output but with large fluctuations.

Full-Wave Rectifier

- **Circuit:** Center-tapped transformer, two diodes D_1 and D_2 , load resistor R_L .

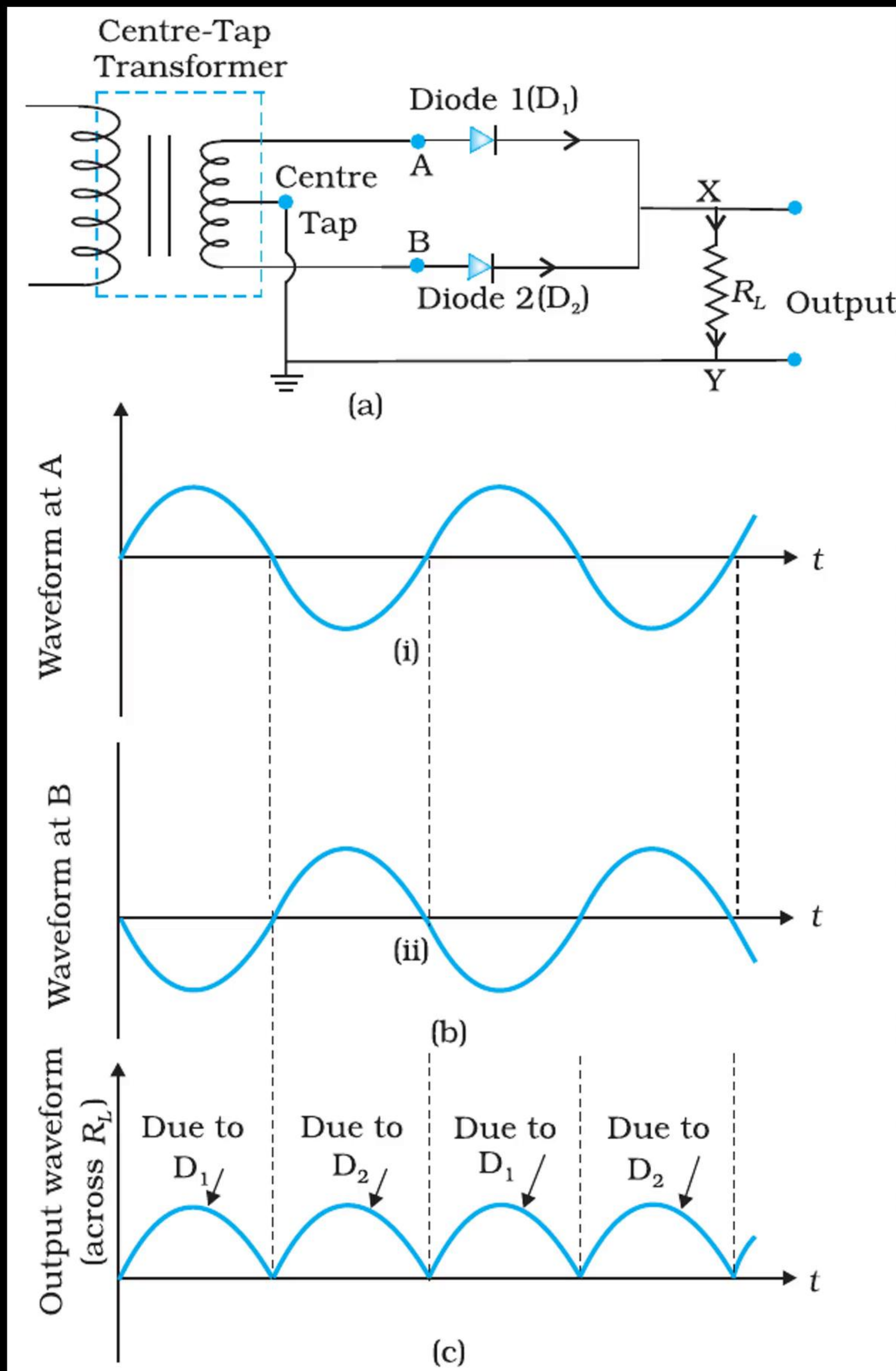
- **Working:**

- Positive half-cycle $\rightarrow D_1$ conducts, D_2 off $\rightarrow$ current through R_L .

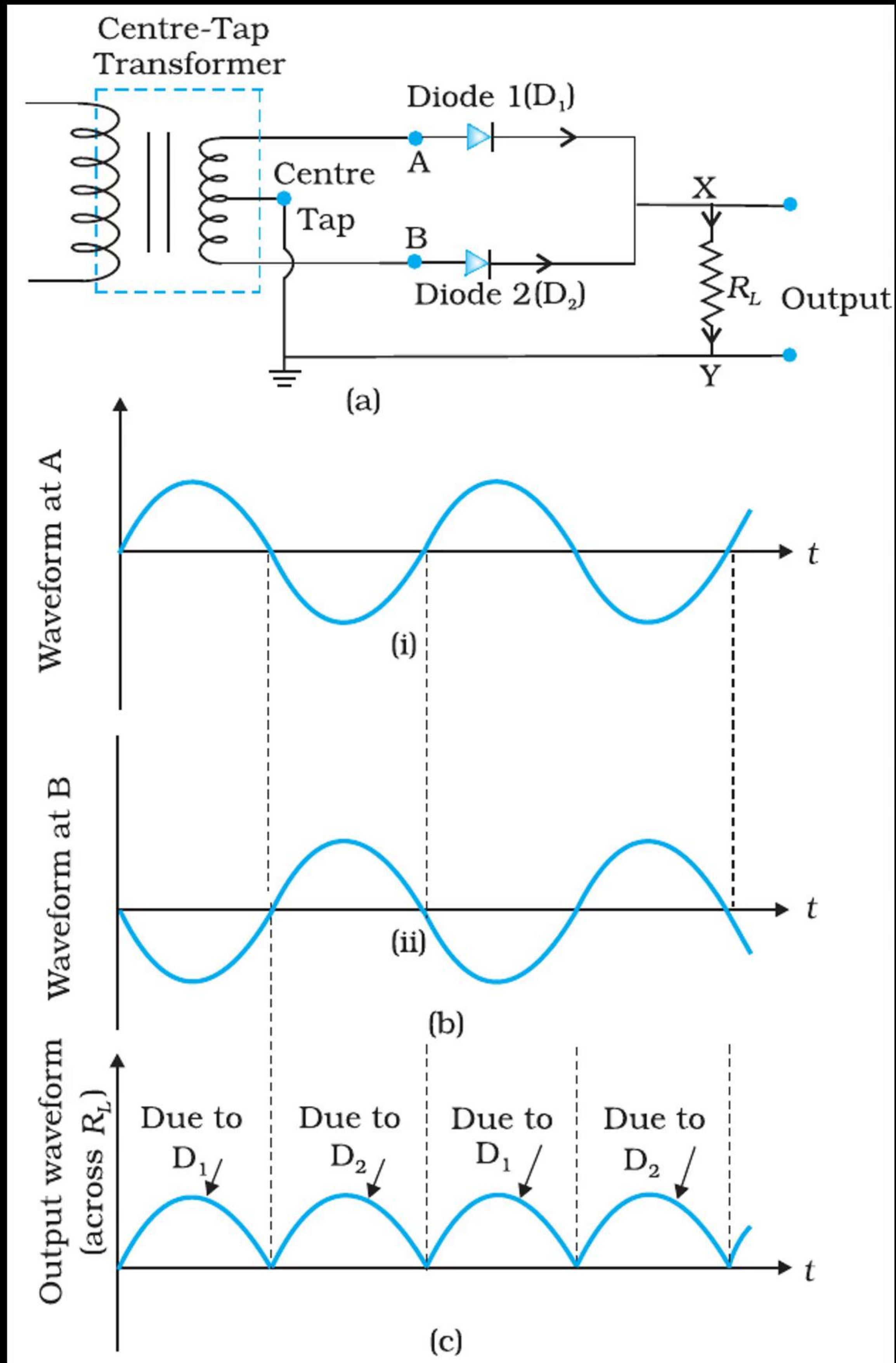
- Negative half-cycle $\rightarrow D_2$ conducts, D_1 off $\rightarrow$ current again in same direction through R_L .

- **Output:** Both halves of AC input appear as positive $\rightarrow$ **pulsating DC**.

- Frequency of output = twice the input AC frequency.

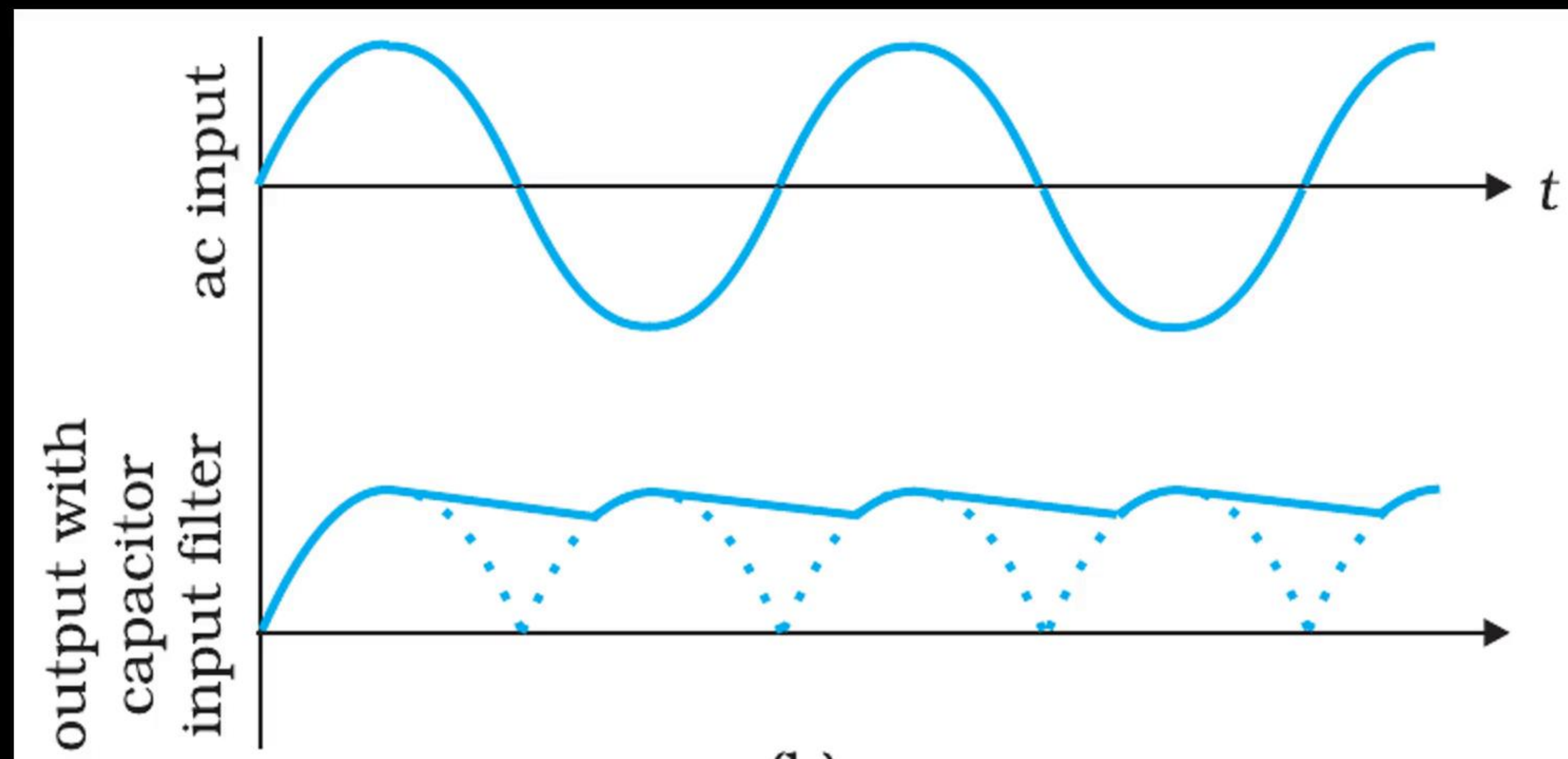
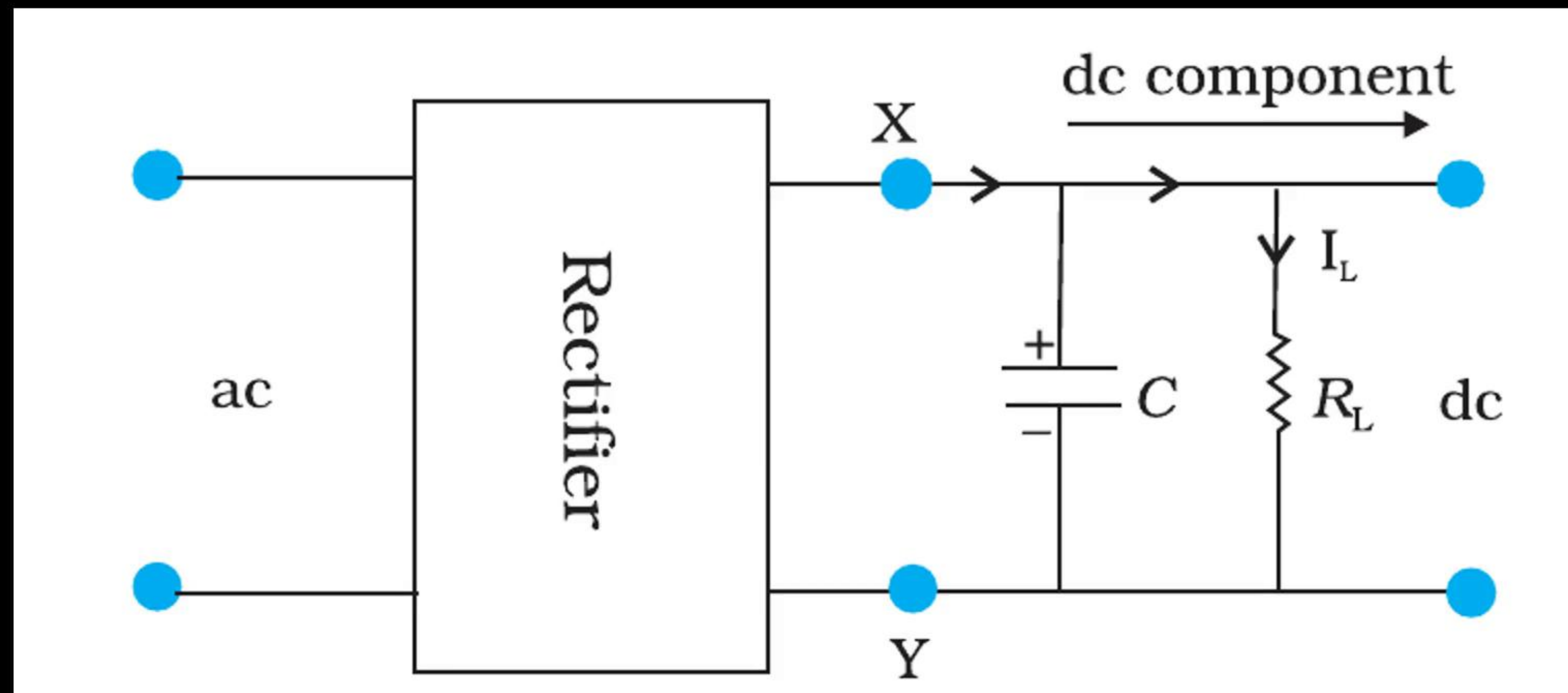


Full-Wave Rectifier: Waveforms



- **Input waveform:** Complete sinusoidal AC.
- **Output waveform:** Both positive and negative half-cycles appear as positive pulses.
- **Output is smoother than half-wave, with higher efficiency.**

Role of Filter Circuits



- Output of rectifier = **pulsating DC**, not pure DC.
- Electronic devices (radios, TV, etc.) require steady DC.
- **Filter circuit:** use large Capacitor connected across R_L .
 - During conduction: capacitor charges.
 - During non-conduction: capacitor discharges slowly through R_L .
- **Result:** Reduced fluctuations, smoother DC output.